

MUSCULAR SYSTEM

Skeletal Muscle tissue

General Functions of muscle(skeletal, smooth, cardiac)

- The major functions of muscles include:
 - 1. ***Body movement.***
 - Contraction of skeletal muscles is responsible for the overall movements of the body, such as walking, running, or manipulating objects with the hands.
 - 2. ***Maintenance of posture.***
 - Skeletal muscles constantly maintain tone, which keeps us sitting or standing erect.
 - 3. ***Respiration.***
 - Skeletal muscles of the thorax are responsible for the movements necessary for respiration.
 - 4. ***Production of body heat.***
 - When skeletal muscles contract, heat is given off as a by-product.
 - This released heat is critical to the maintenance of body temperature.

Functions of muscle tissue

□ 5. ***Communication.***

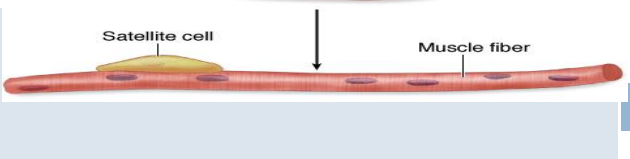
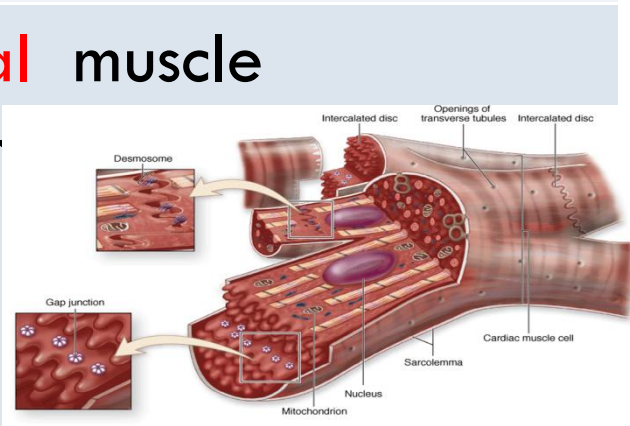
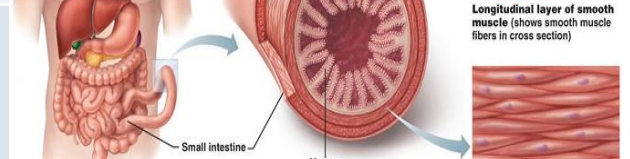
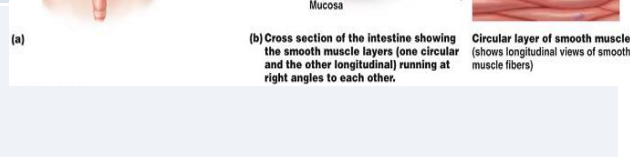
- Skeletal muscles are involved in all aspects of communication, such as **speaking, writing, typing, gesturing, and facial expression.**

□ 6. ***Constriction of organs and vessels.***

- The contraction of smooth muscle within the walls of internal organs and vessels causes constriction of those structures.
- This constriction can help propel and mix food and water in the digestive tract, propel secretions from organs, and regulate blood flow through blood vessels.

□ 7. ***Heartbeat.***

- The contraction of cardiac muscle causes the heart to beat, propelling blood to all parts of the body.

General Features	Description	
Cellularity	Highly cellular	
Extracellular matrix	Moderate	
Proliferation	Moderate for skeletal muscle Limited for cardiac muscle High for smooth muscle	
Polarity	Poor	
vasculature	Highly vascular	
Innervation	Highly innervated	

Functional features of muscle tissue

- There are four characteristics associated with muscle tissue:
 - Excitability
 - Tissue can receive & respond to stimulation
 - Contractility
 - Tissue can shorten & thicken
 - Extensibility
 - Tissue can lengthen
 - Elasticity
 - After contracting or lengthening, tissue always wants to return to its resting state

Types of muscle tissue cells



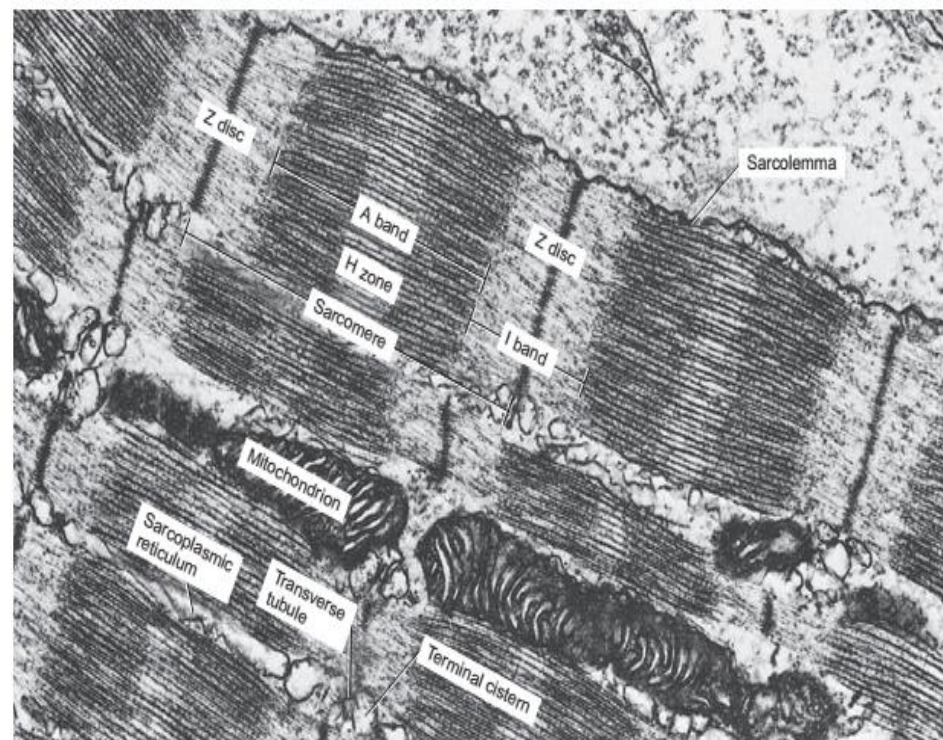
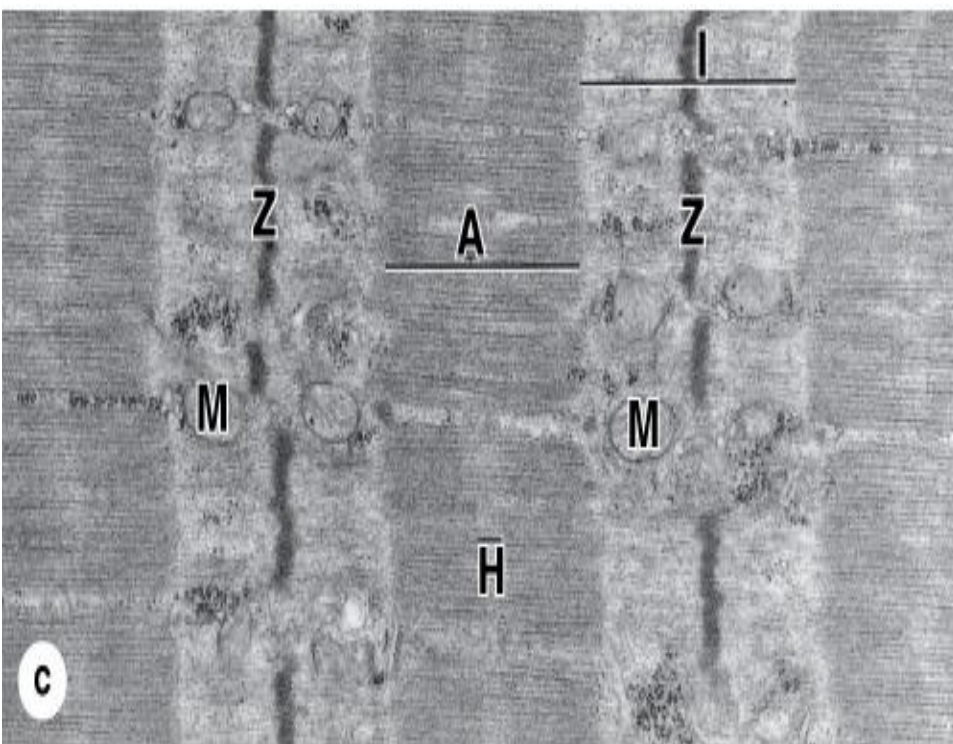
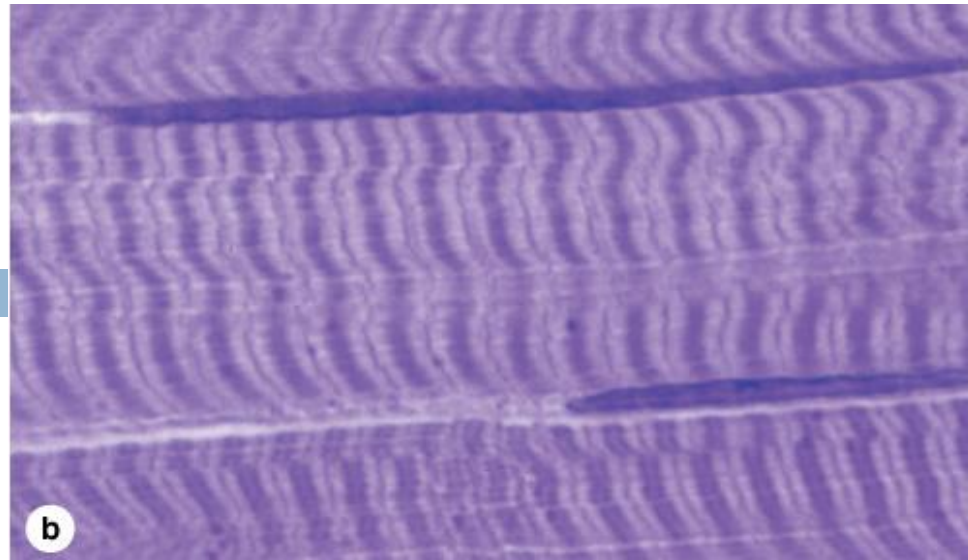
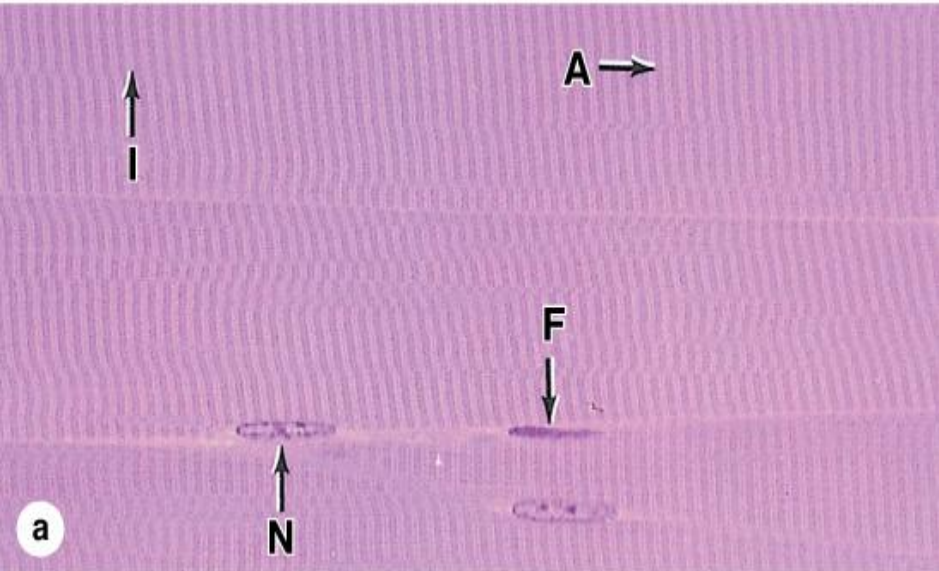
- There are three types of muscle tissue cells:
 - ▣ **Skeletal muscle cells**
 - ▣ **Cardiac muscle cells**
 - ▣ **Smooth muscle cells**

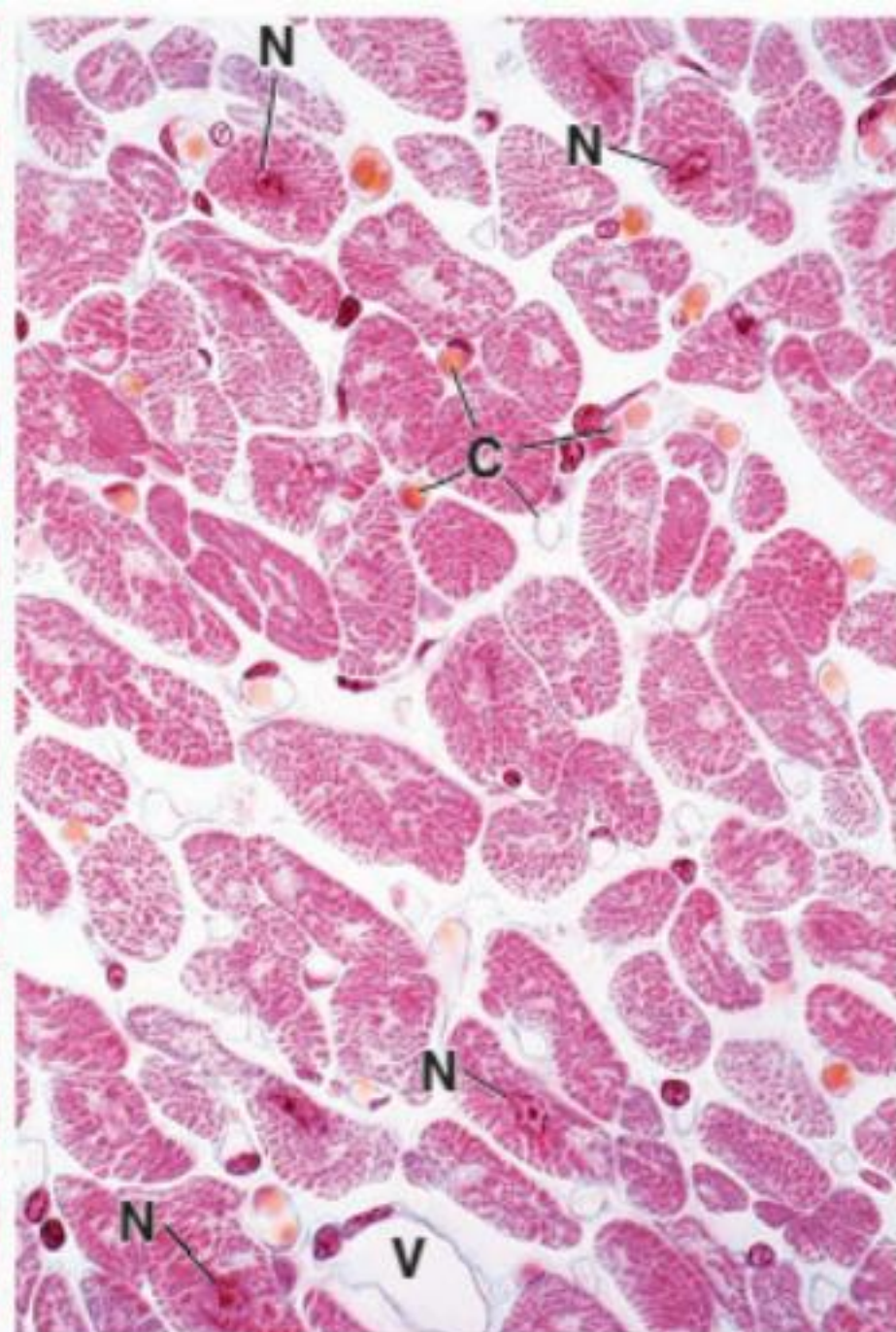
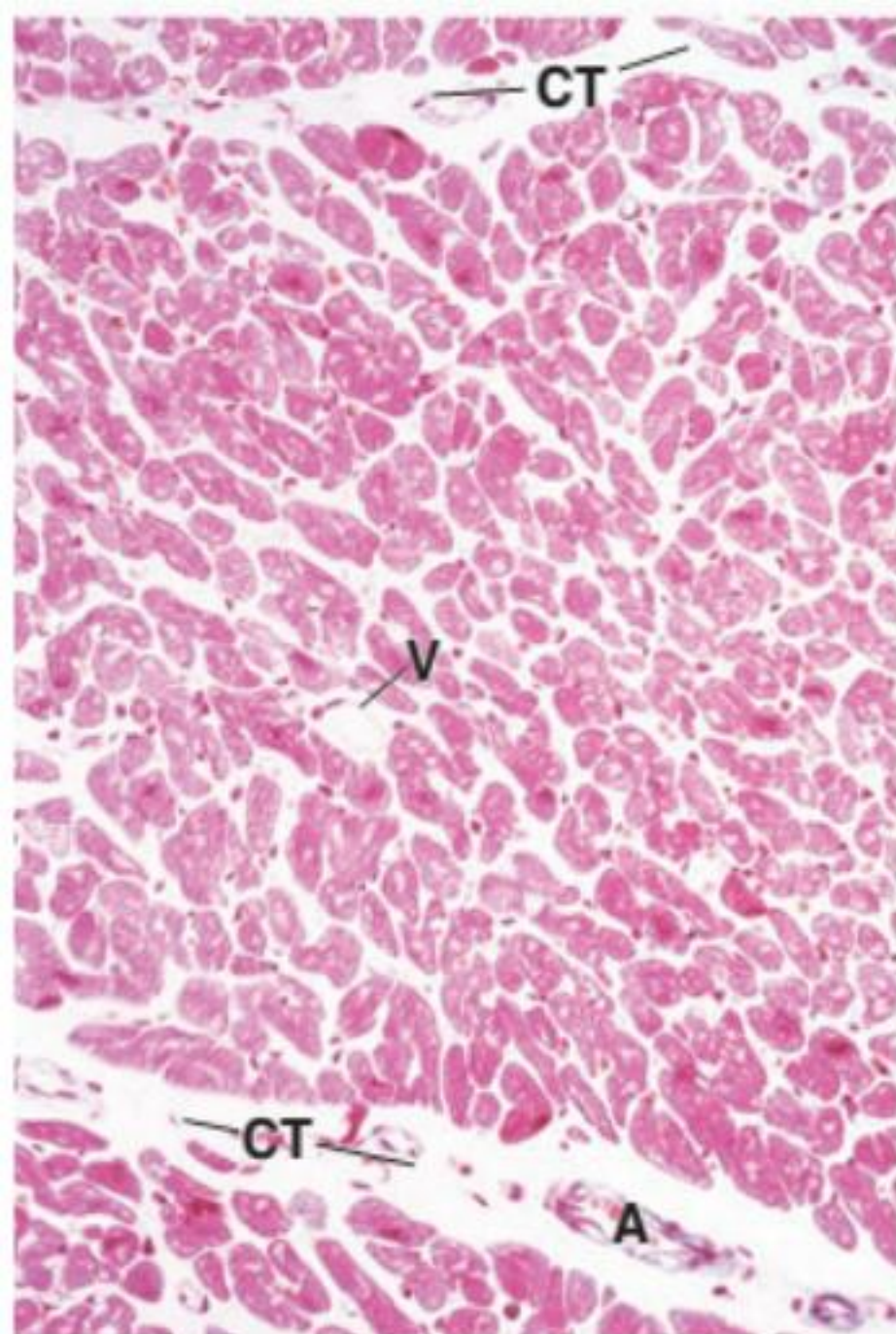
There are three types of muscle tissue: skeletal, smooth, and cardiac.

Features	Skeletal Muscle	Smooth Muscle	Cardiac Muscle
Location	Attached to bones	Walls of hollow organs, blood vessels, eyes, glands, and skin	Heart
Appearance			
Cell shape	Very long and cylindrical (1 mm–4 cm); extends the length of muscle fascicles, which in some cases is the length of the muscle	Spindle-shaped (15–200 μm in length, 5–8 μm in diameter)	Cylindrical and branched (100–500 μm in length, 12–20 μm in diameter)
Nucleus	Multiple, peripherally located	Single, centrally located	Single, centrally located
Special cell–cell attachments	None	Gap junctions join some visceral smooth muscle cells together	Intercalated disks join cells to one another
Striations	Yes	No	Yes
Control	Voluntary and involuntary (reflexes)	Involuntary	Involuntary
Capable of spontaneous contraction	No	Yes (some smooth muscle)	Yes
Function	Body movement	Food movement through the digestive tract, emptying of the urinary bladder, regulation of blood vessel diameter, change in pupil size, contraction of many gland ducts, movement of hair, and many other functions	Pumps blood; contractions provide the major force for propelling blood through blood vessels

T tubules and SR	Has triads at A–I junctions	Has caveolae (but no T tubules) and some SER	Has dyads at Z disks
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Sarcomere	Yes	No	Yes
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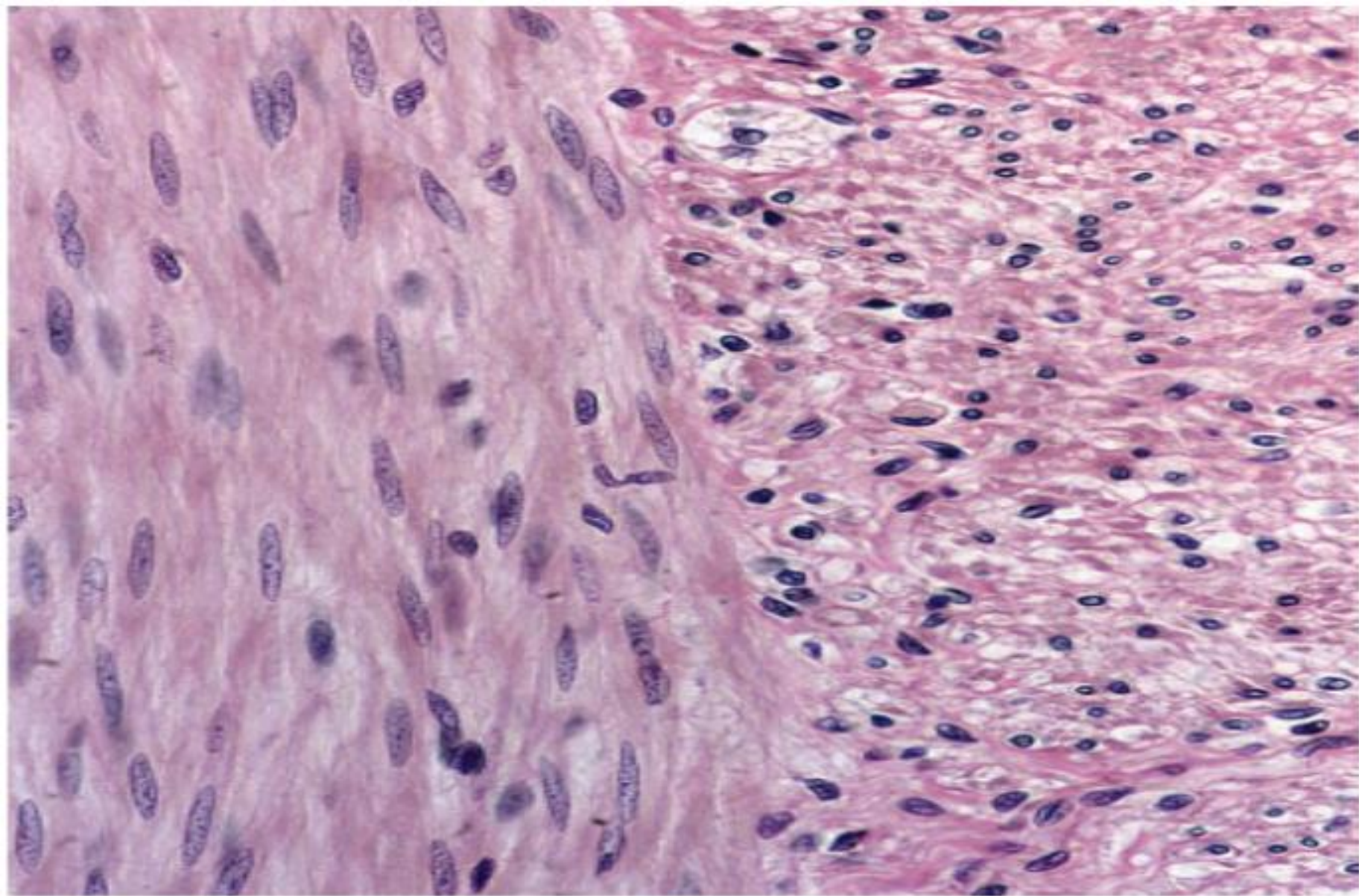
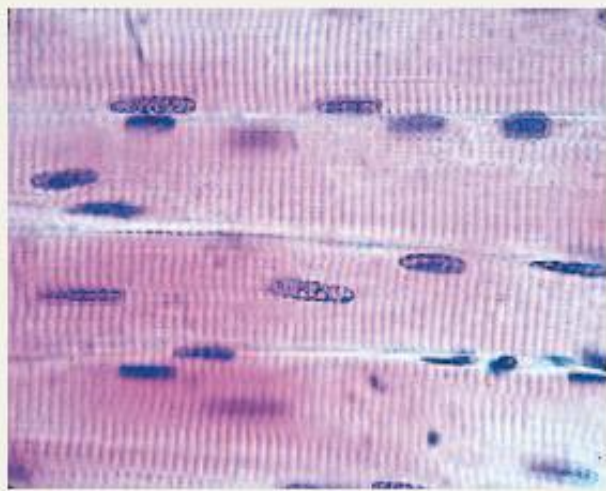
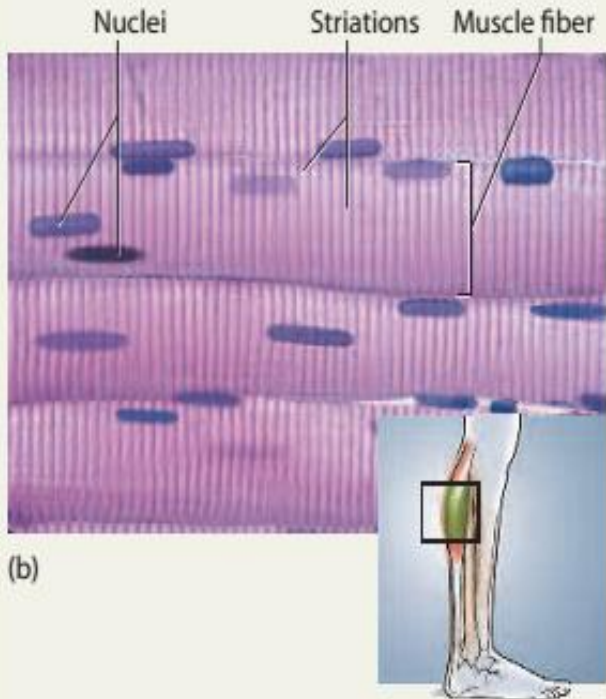


FIGURE 11.18 • Photomicrograph of smooth muscle from a human colon. The smooth muscle shown in this micrograph is arranged in two layers. On the left, the muscle cells are cut in

Skeletal Muscle

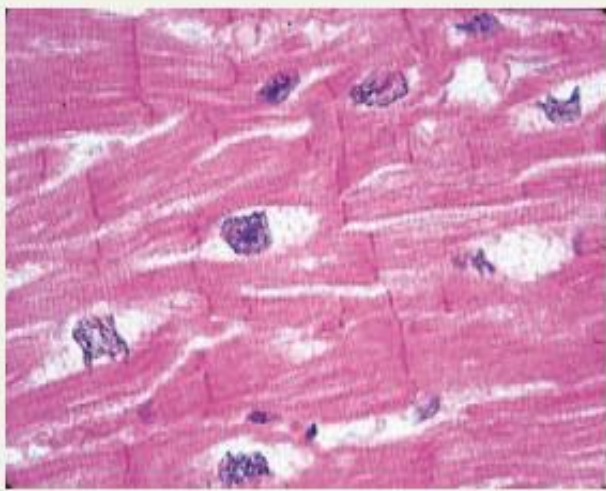


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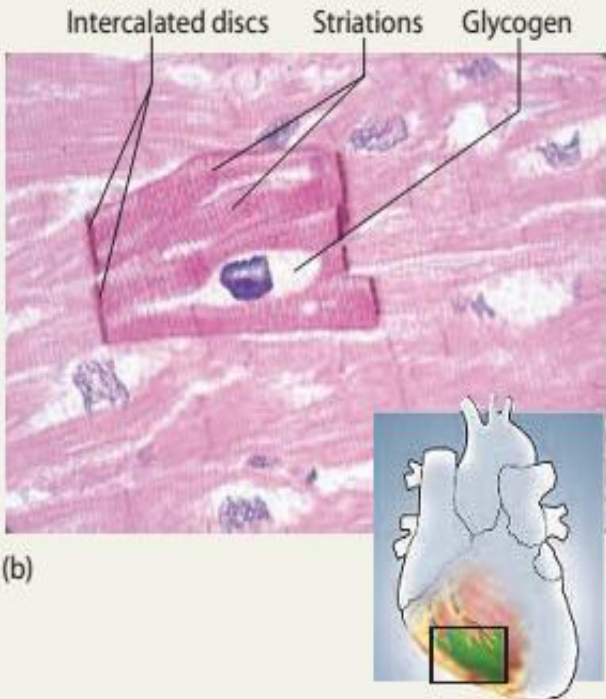


(b)

Cardiac Muscle

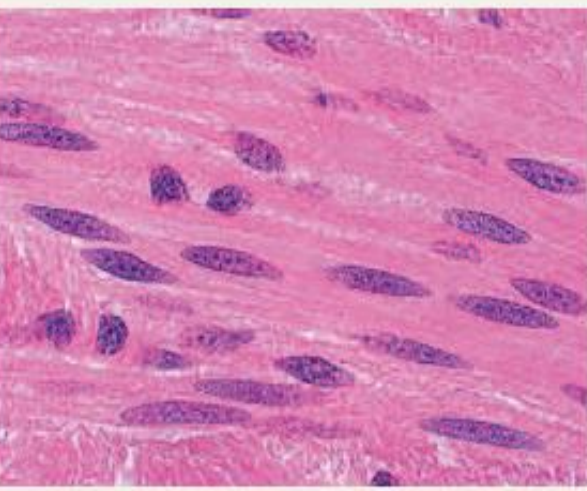


(a)



(b)

Smooth Muscle



(a)

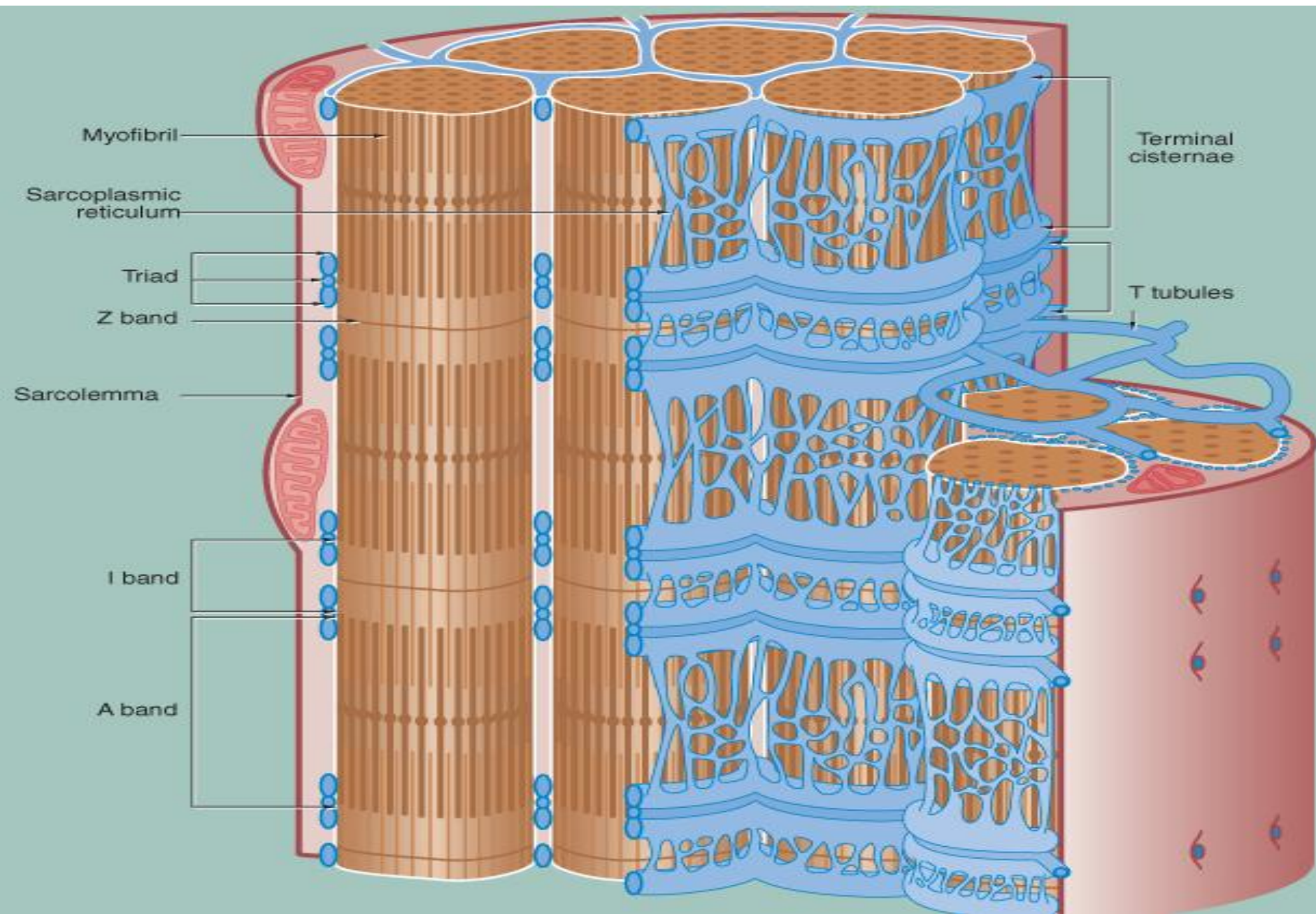


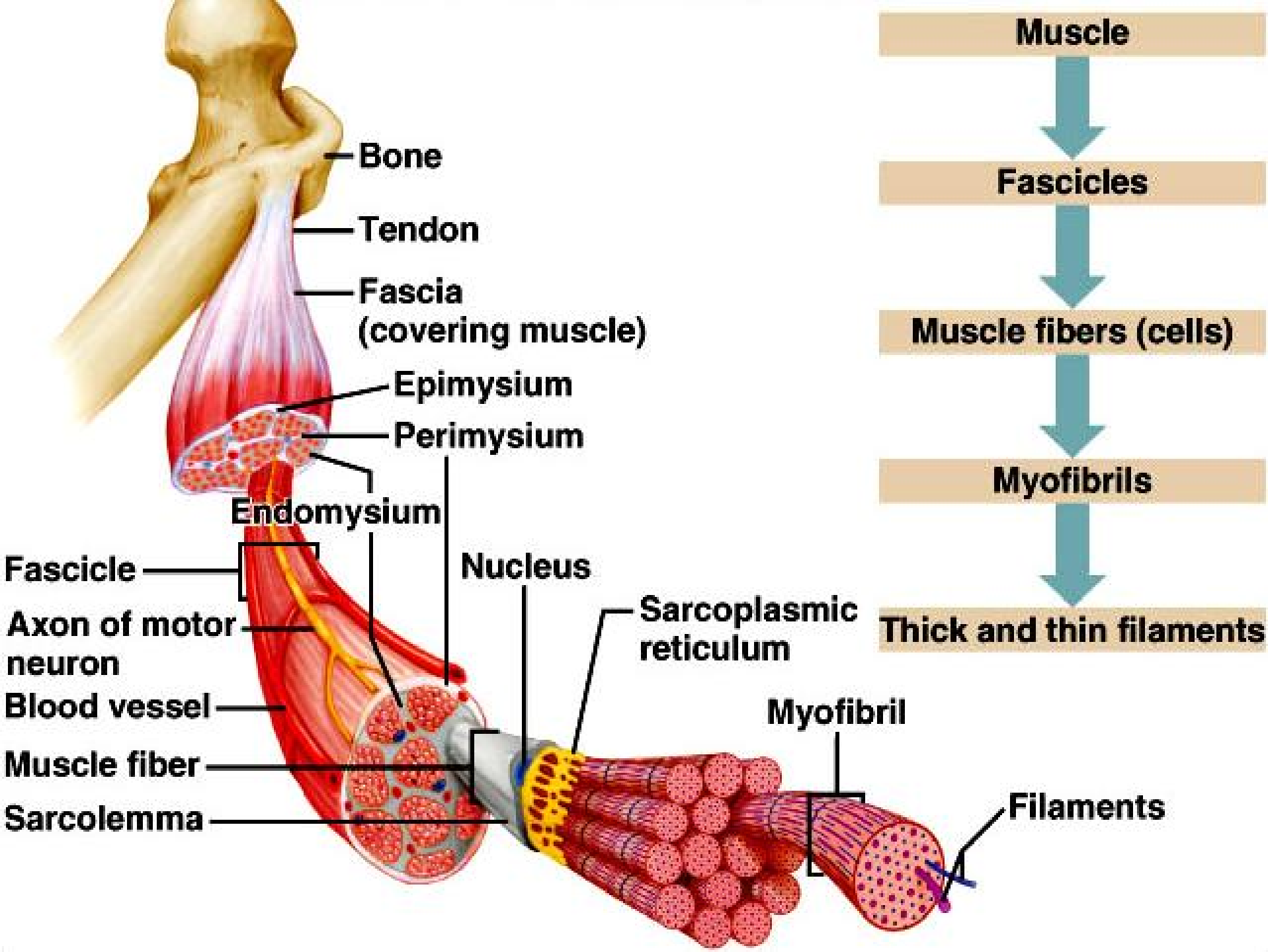
(b)

Common features of muscle tissue



- Organelles of muscle cells
- External lamina + elastic lamina
- Connective tissue investment
- Neurovascular bundles





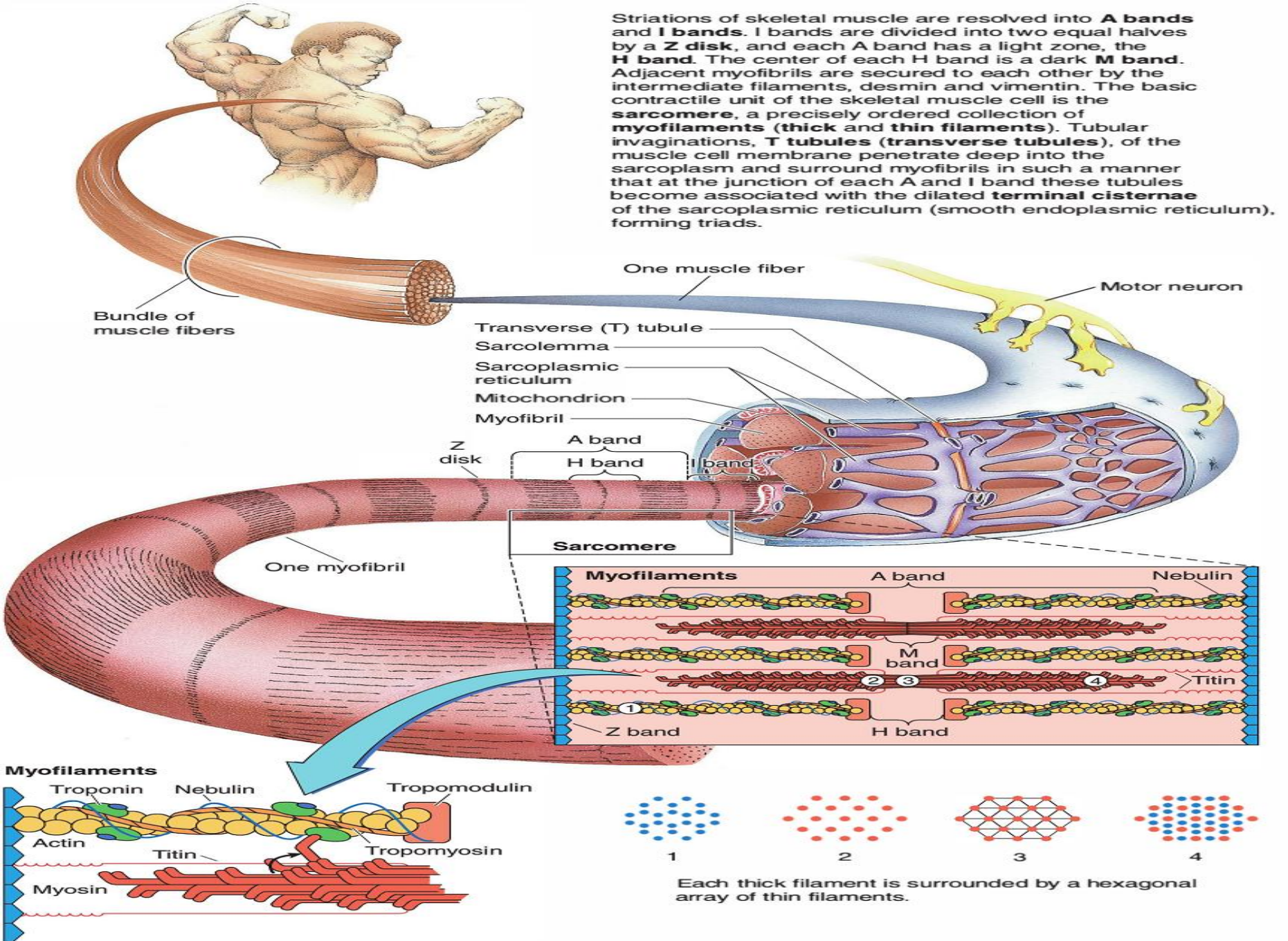
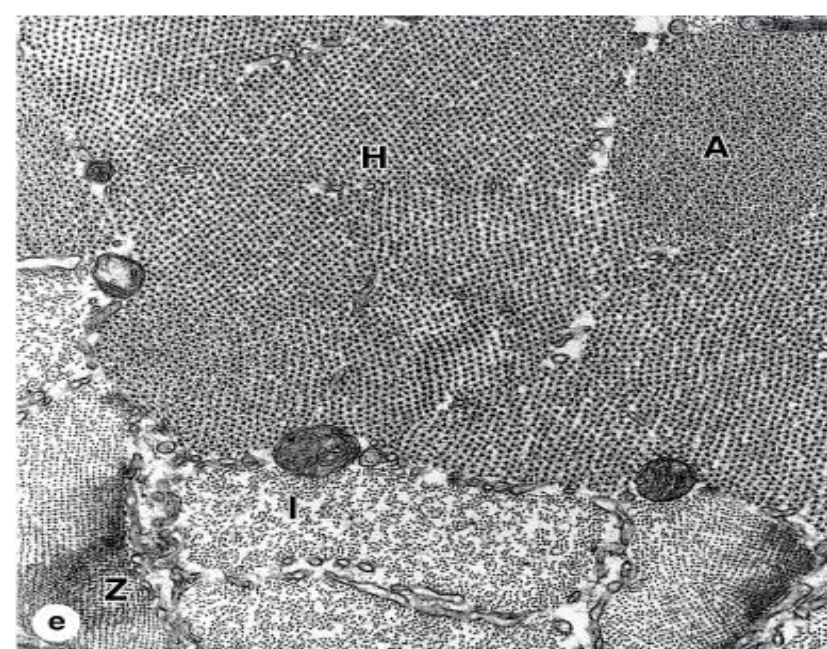
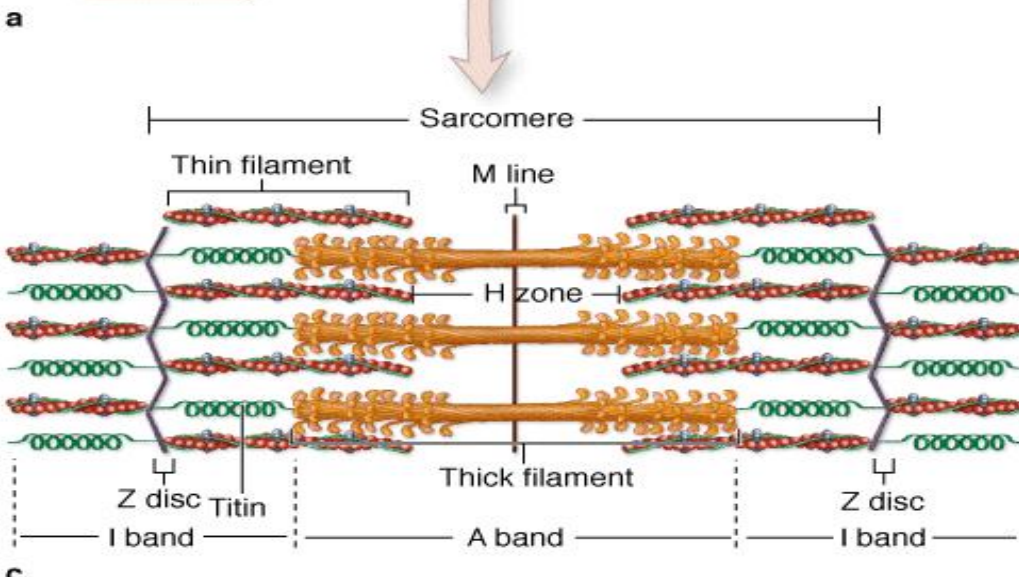
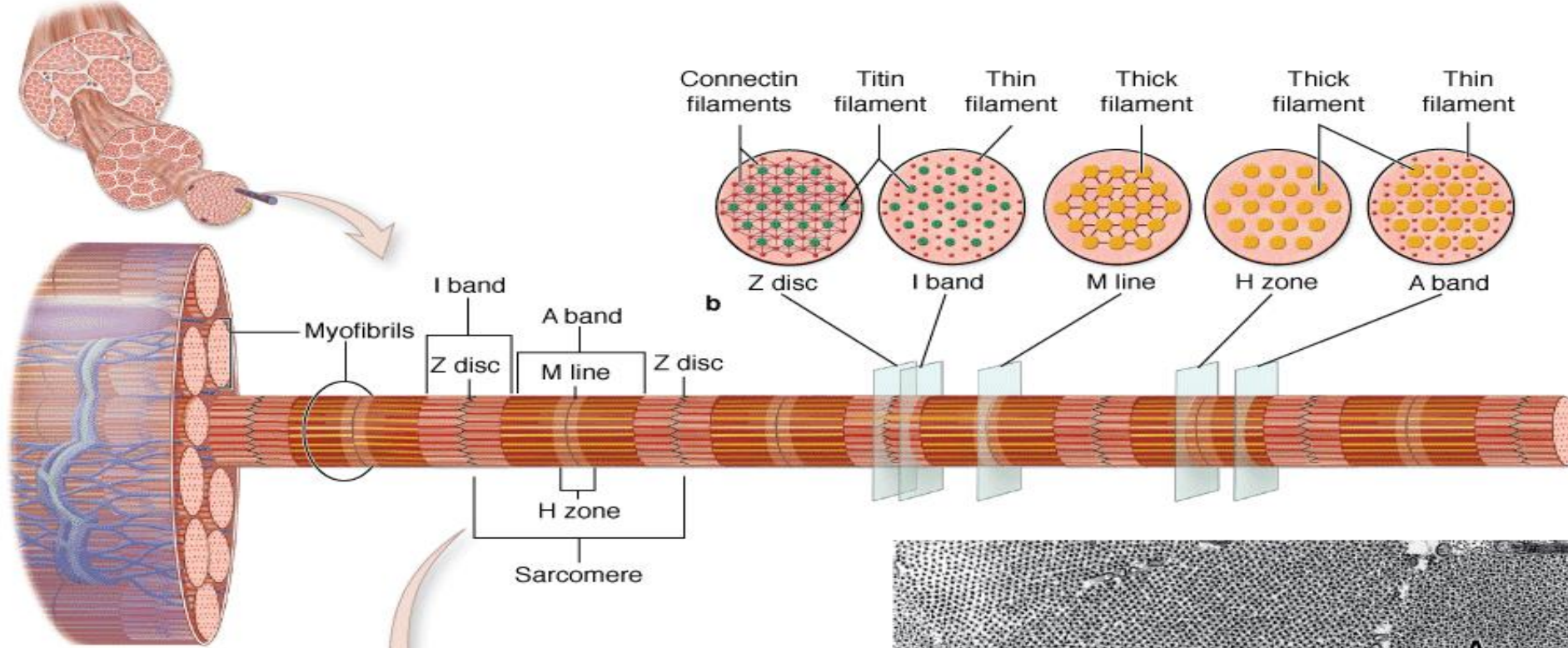
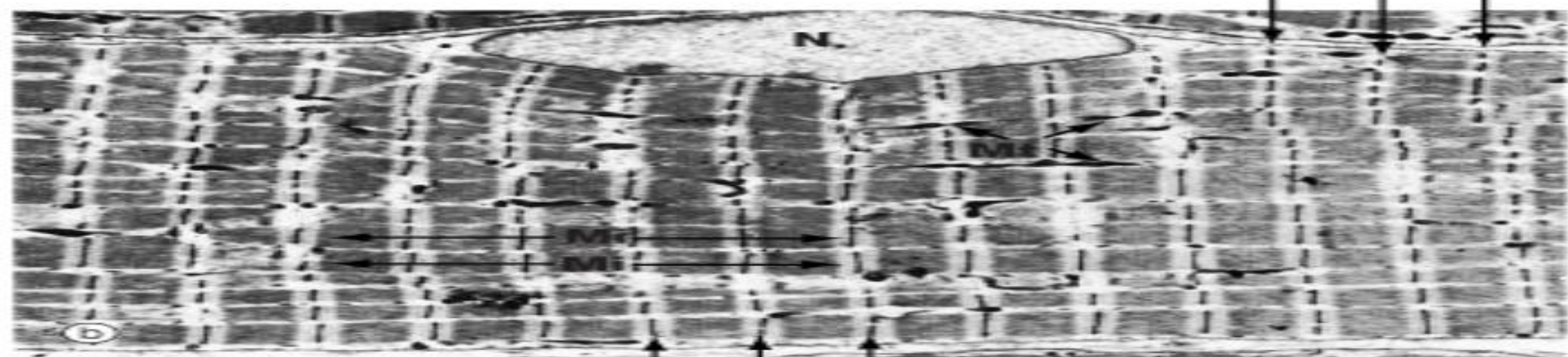
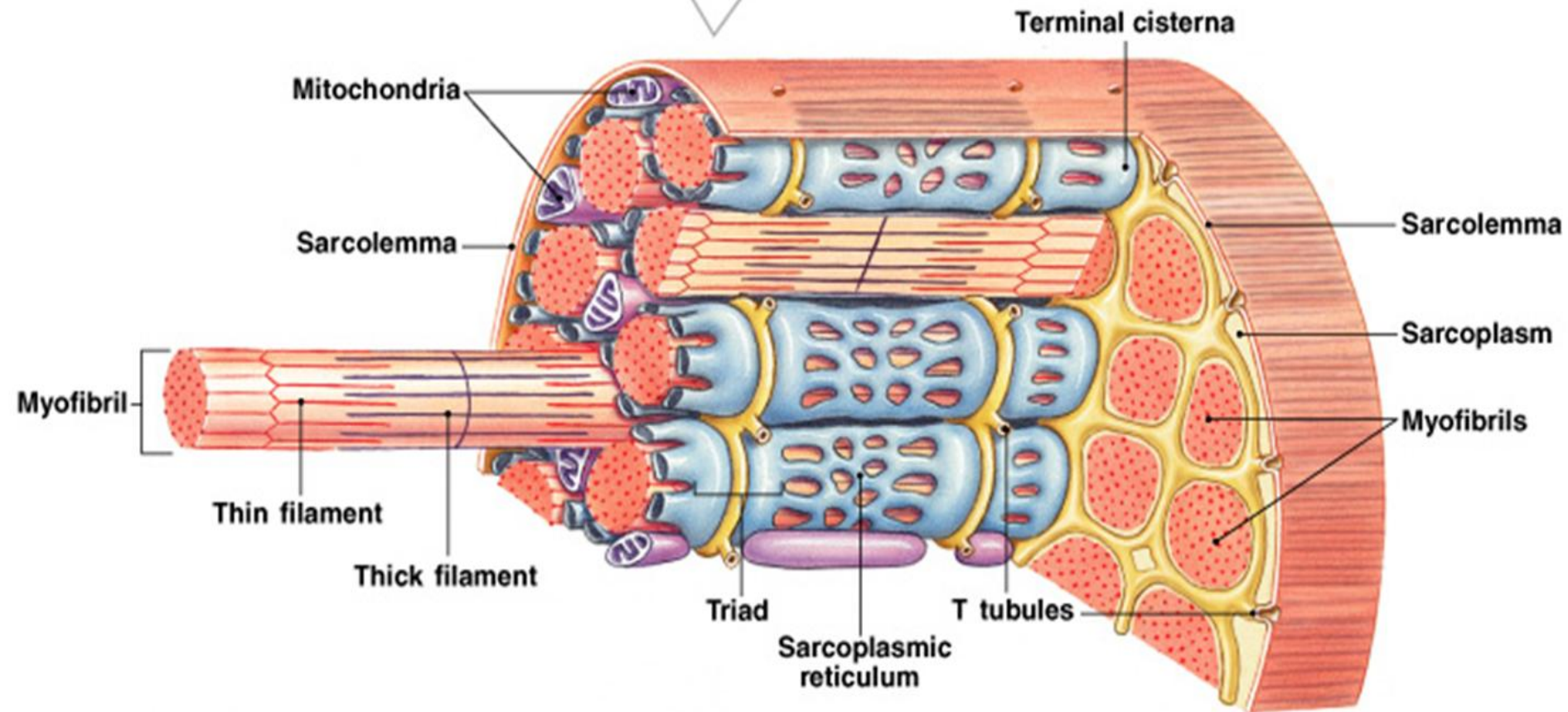
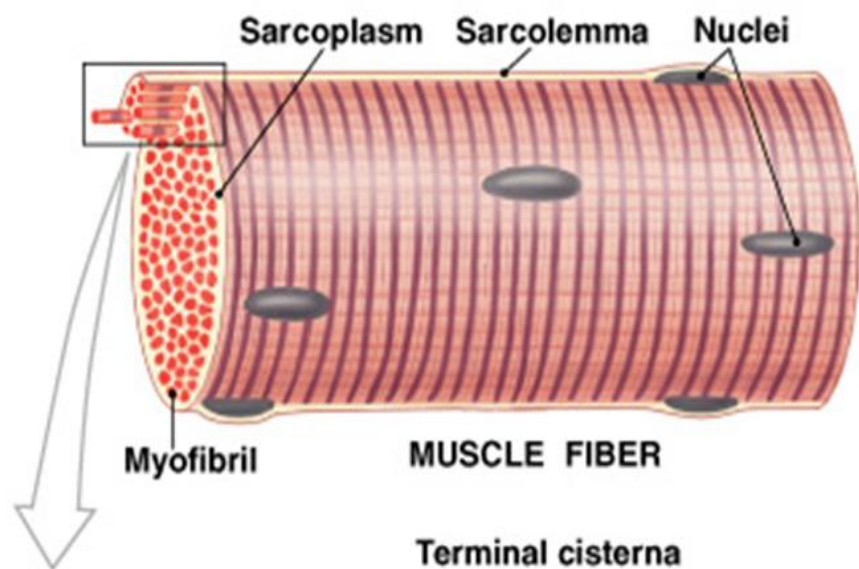
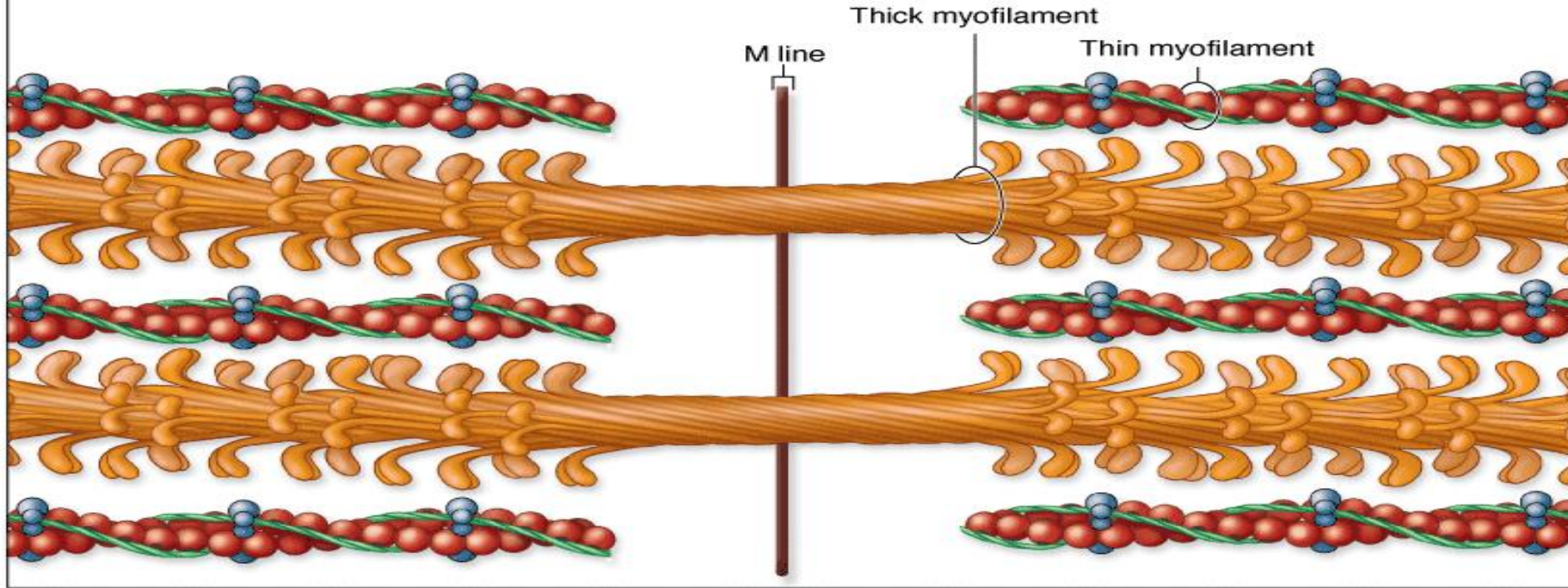


FIGURE 8.1. Diagram of skeletal muscle and its components as observed by light and electron microscopy. (Reprinted with permission from Gartner LP, Hiatt JL. *Color Atlas of Histology*. 5th ed. Baltimore, MD: Lippincott Williams & Wilkins; 2009:116.)

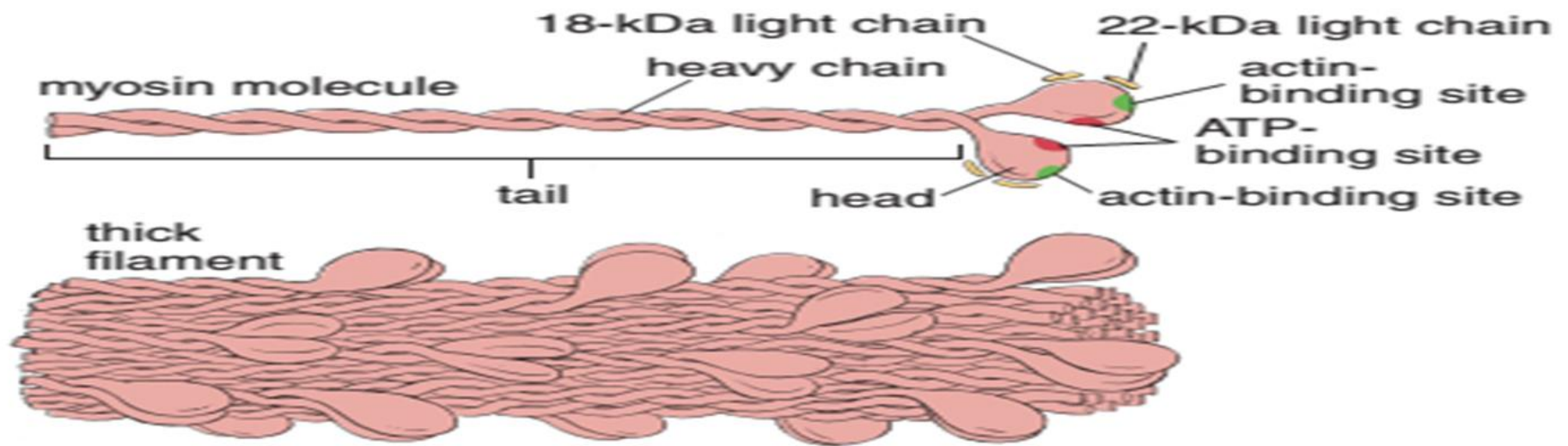




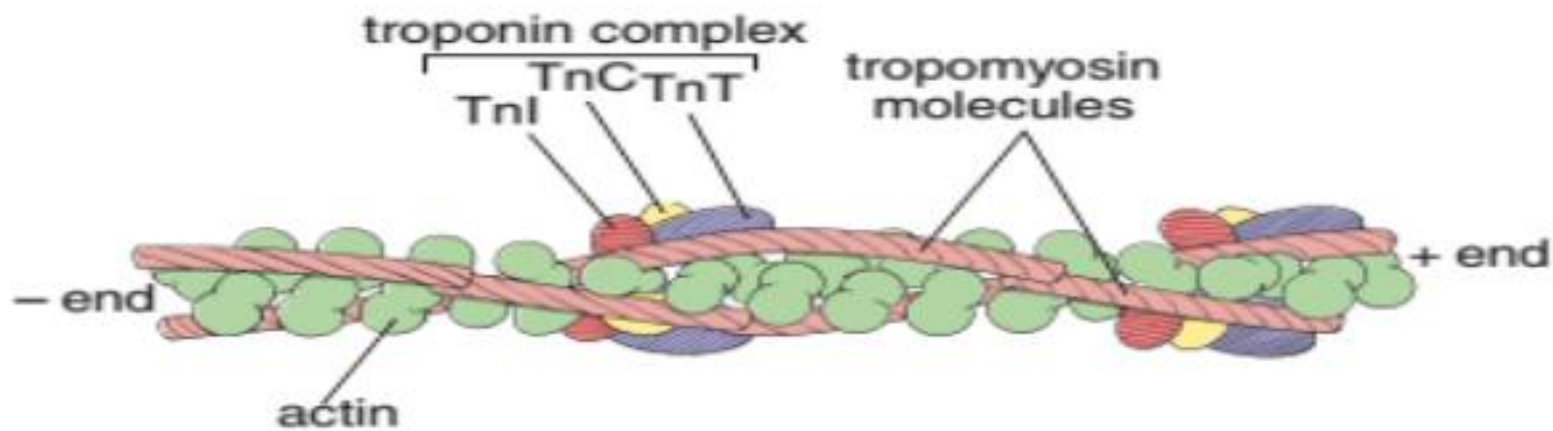
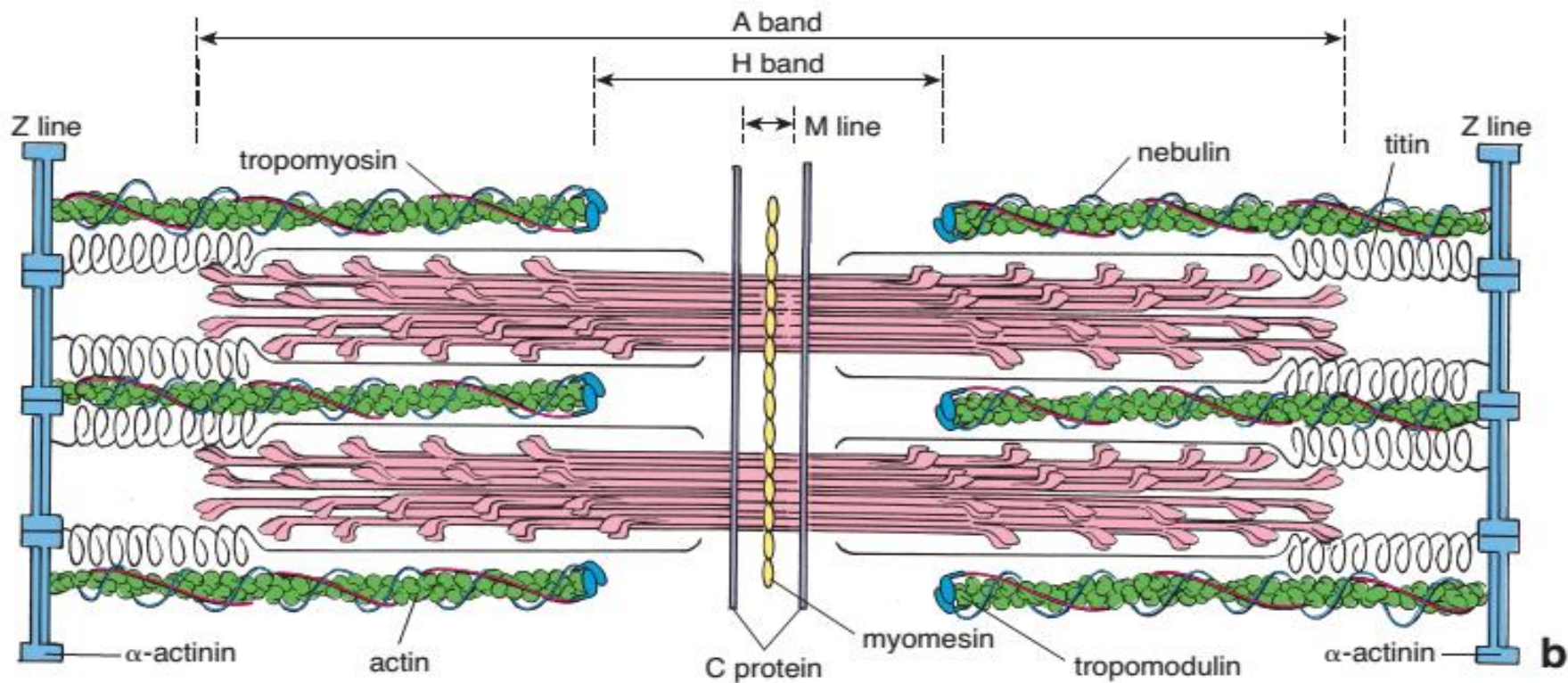


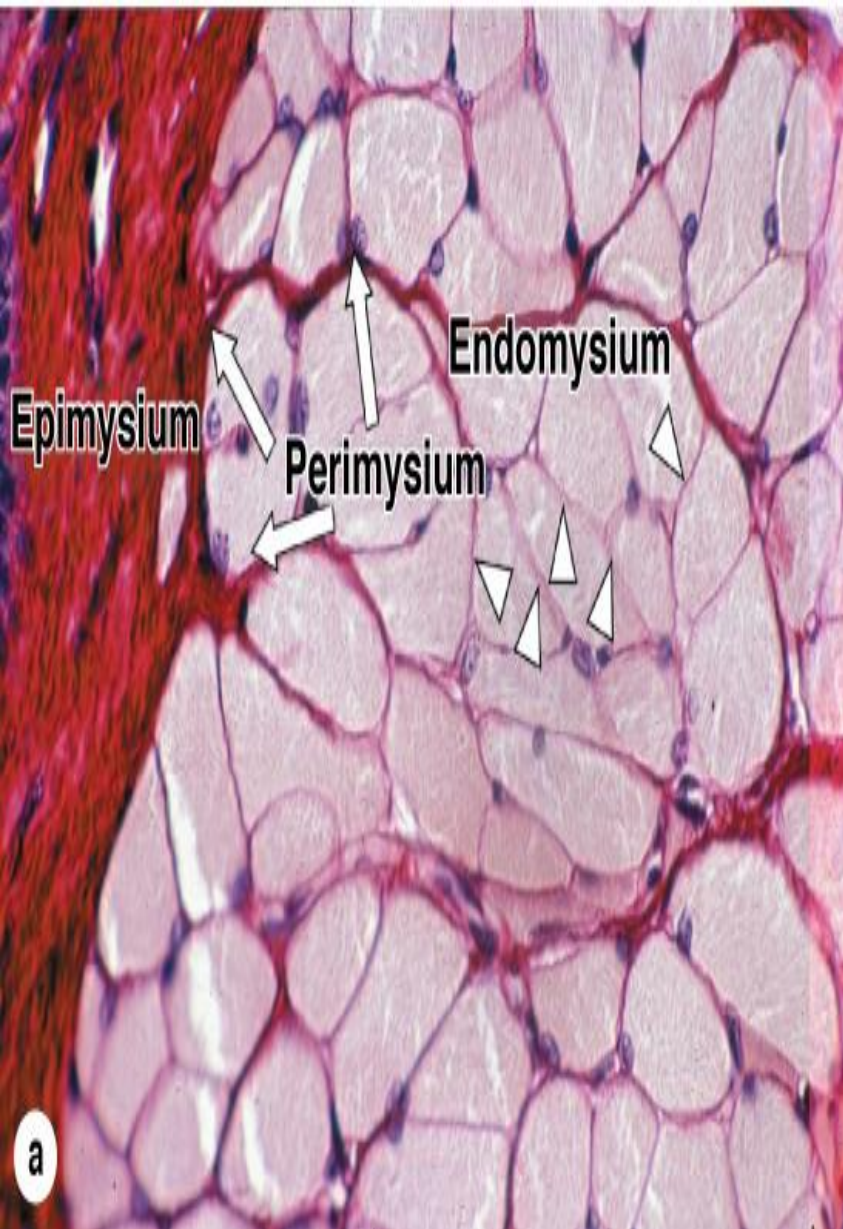


c Comparison of thick and thin filaments

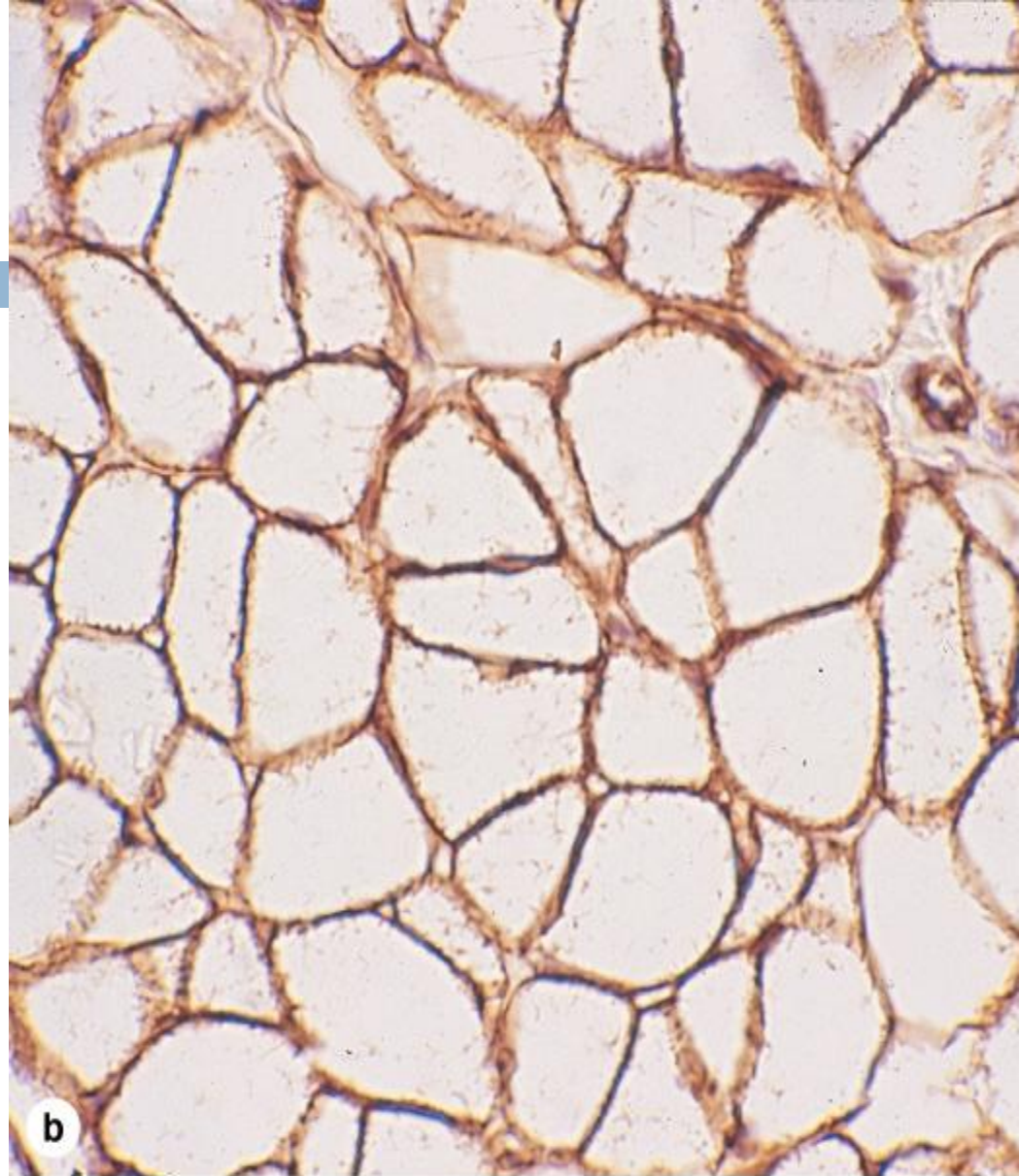


Myosin II



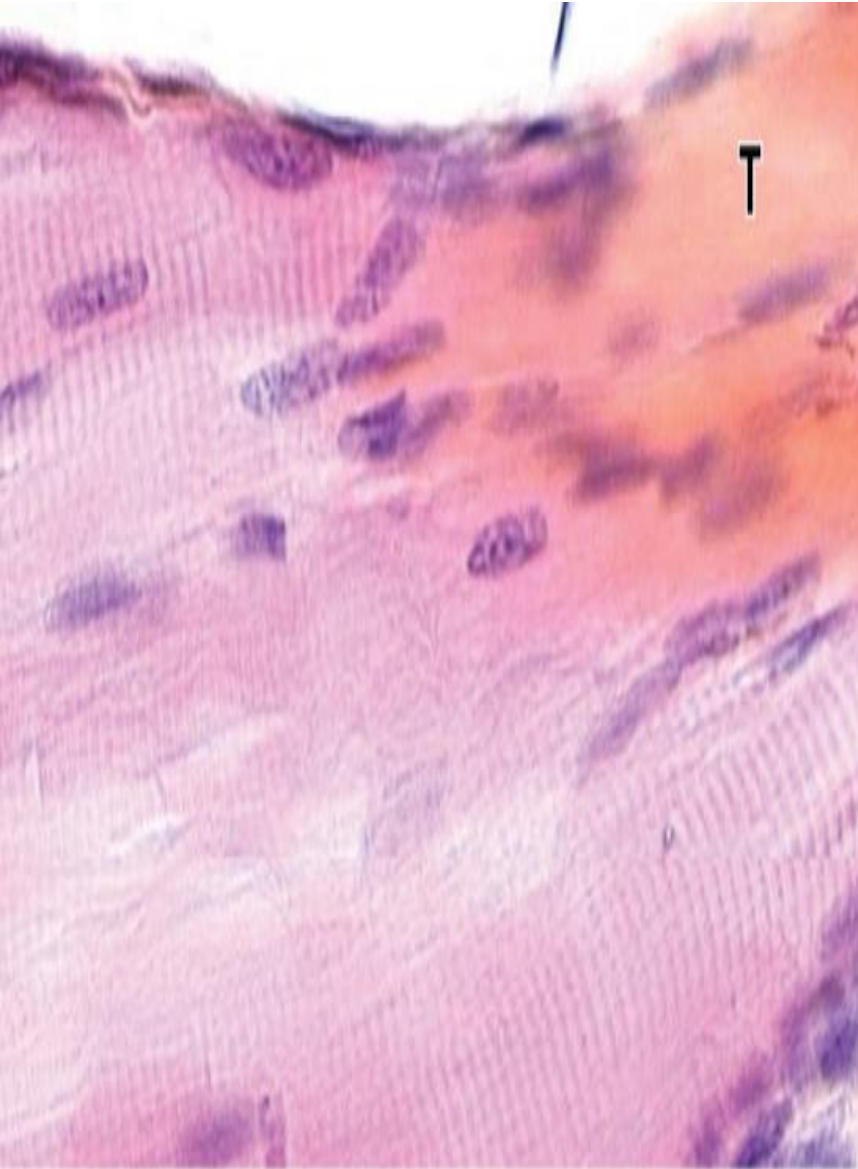


X200. H&E



X400. Anti-laminin

Skeletal muscle. Micrograph shows a cross section of striated muscle demonstrating connective tissue and cell nuclei.



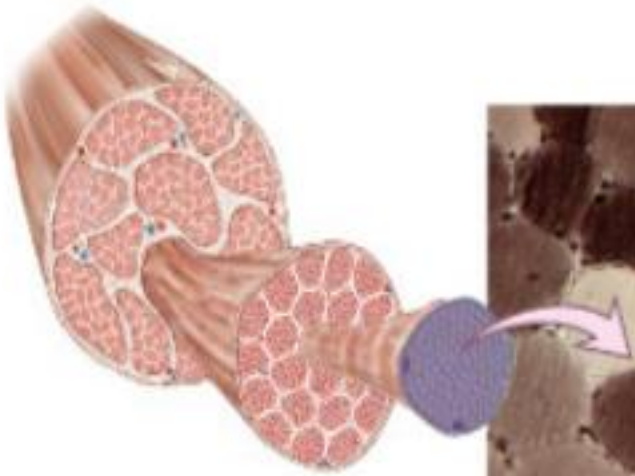
X200. Giemsa with polarized light

Myotendinous junction.

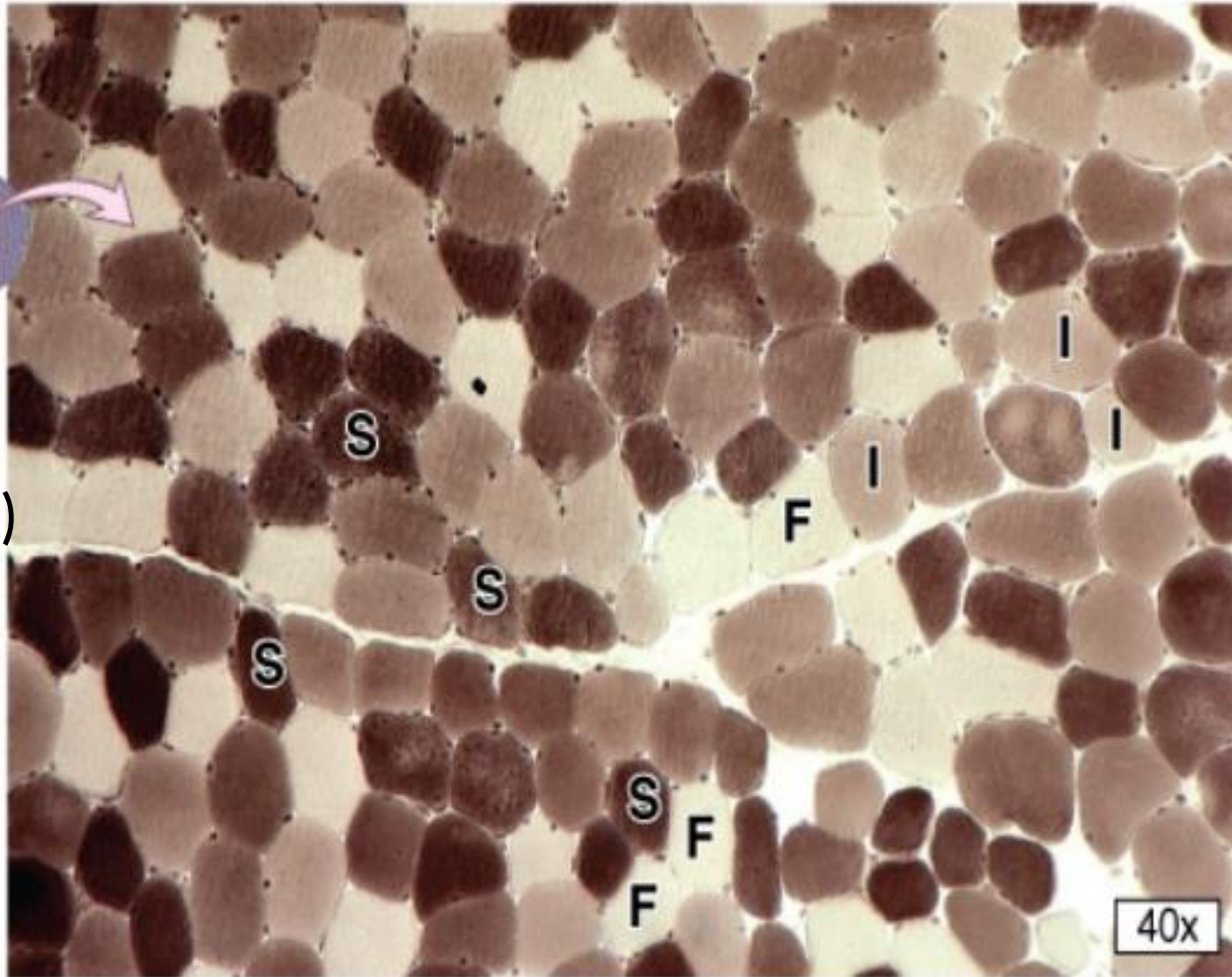
Vasculature of muscle: Capillaries of skeletal muscle.

Table 9.2 Structural and Functional Characteristics of the Three Types of Skeletal Muscle Fibers

	SLOW OXIDATIVE FIBERS	FAST OXIDATIVE FIBERS	FAST GLYCOLYTIC FIBERS
Metabolic Characteristics	Oxidative fibers	Oxidative-glycolytic fibers	Glycolytic fibers
Speed of contraction	Slow	Fast	Fast
Myosin ATPase activity	Slow <i>,Low in ATPase</i>	Fast <i>,Intermediate in ATPase</i>	Fast <i>,High in ATPase & Phosphorylases</i>
Enzyme content Primary pathway for ATP synthesis	High Oxidative Enzymes Aerobic	Intermediate Oxidative Enzymes Aerobic (some anaerobic glycolysis)	Low in Oxidative Enzymes Anaerobic glycolysis
Myoglobin content	High	High	Low
Glycogen stores	Low	Intermediate	High
Recruitment order	First	Second	Third
Rate of fatigue	Slow (fatigue-resistant)	Intermediate (moderately fatigue-resistant)	Fast (fatigable)
Activities Best Suited For			
	Endurance-type activities—e.g., running a marathon; maintaining posture (antigravity muscles)	Sprinting, walking	Short-term intense or powerful movements, e.g., hitting a baseball
Structural Characteristics			
Color	Red	Red to pink	White (pale)
Fiber diameter	Small	Intermediate	Large
Mitochondria	Many	Many	Few
Capillaries	Many	Many	Few



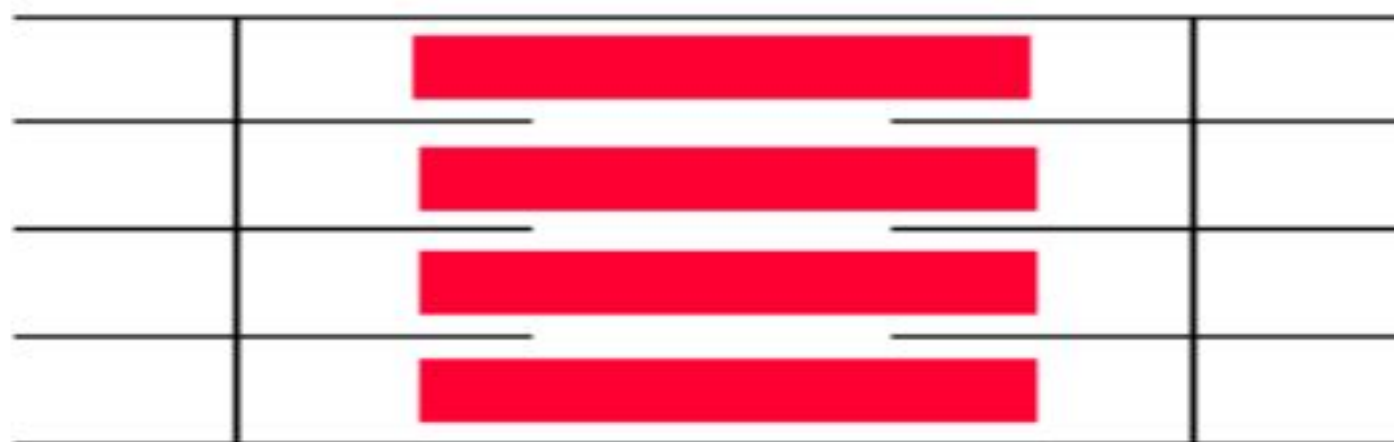
- Slow (S)
- Intermediate (I)
- Fast (F)



Cross-section of skeletal muscle **stained histochemically** to detect the density of myofibrillar myosin-ATPase . This staining can be used to demonstrate the distribution of slow (S) type I fibers, intermediate (I) type IIa fibers, and fast (F) type IIb fibers.

Effects of Contraction on Skeletal Muscle Cross-Bands

Bands	Myofilament Component	Change in Bands During Contraction
I	Thin only	Shorten
H	Thick only	Shorten
A	Thick and thin	N change in length
Z disks	Thin only (attached by α -actinin)	Move closer together



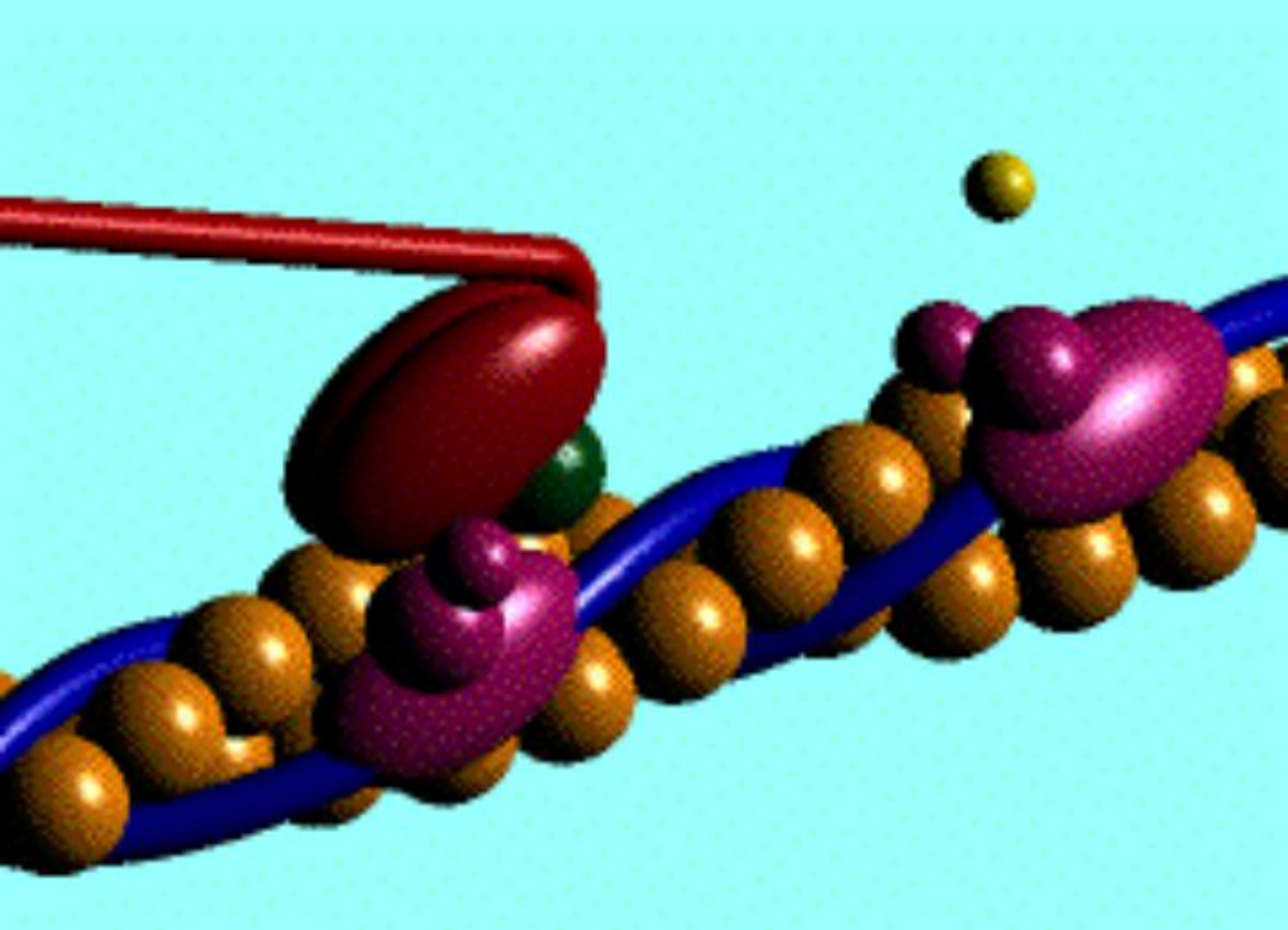
Myosin

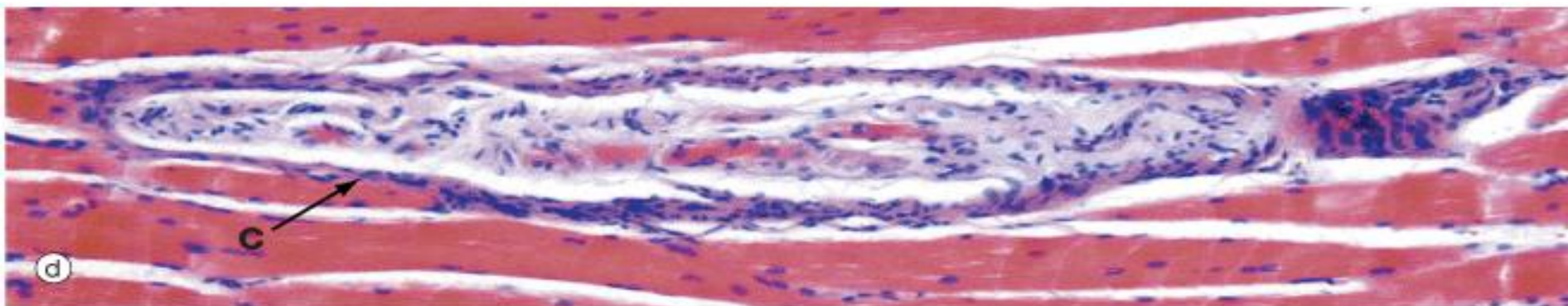
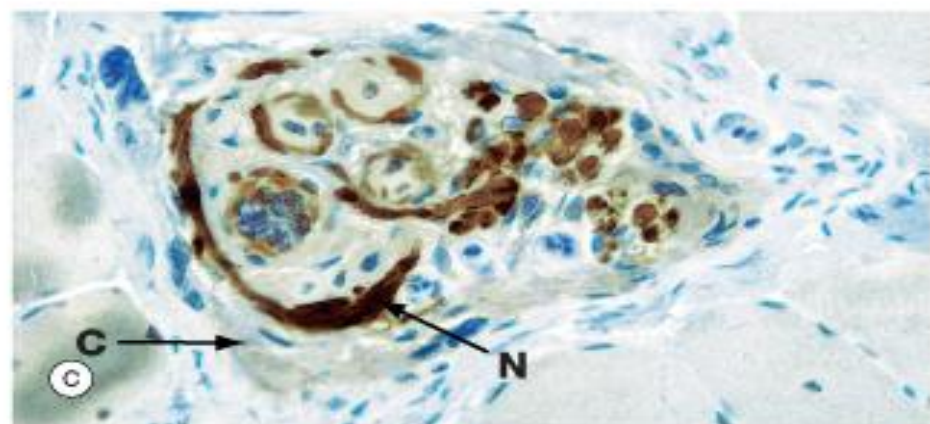
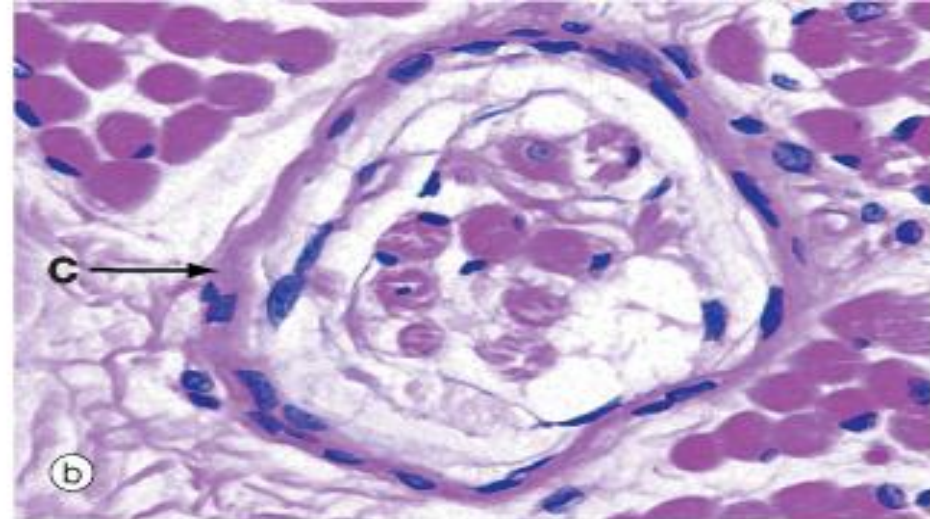
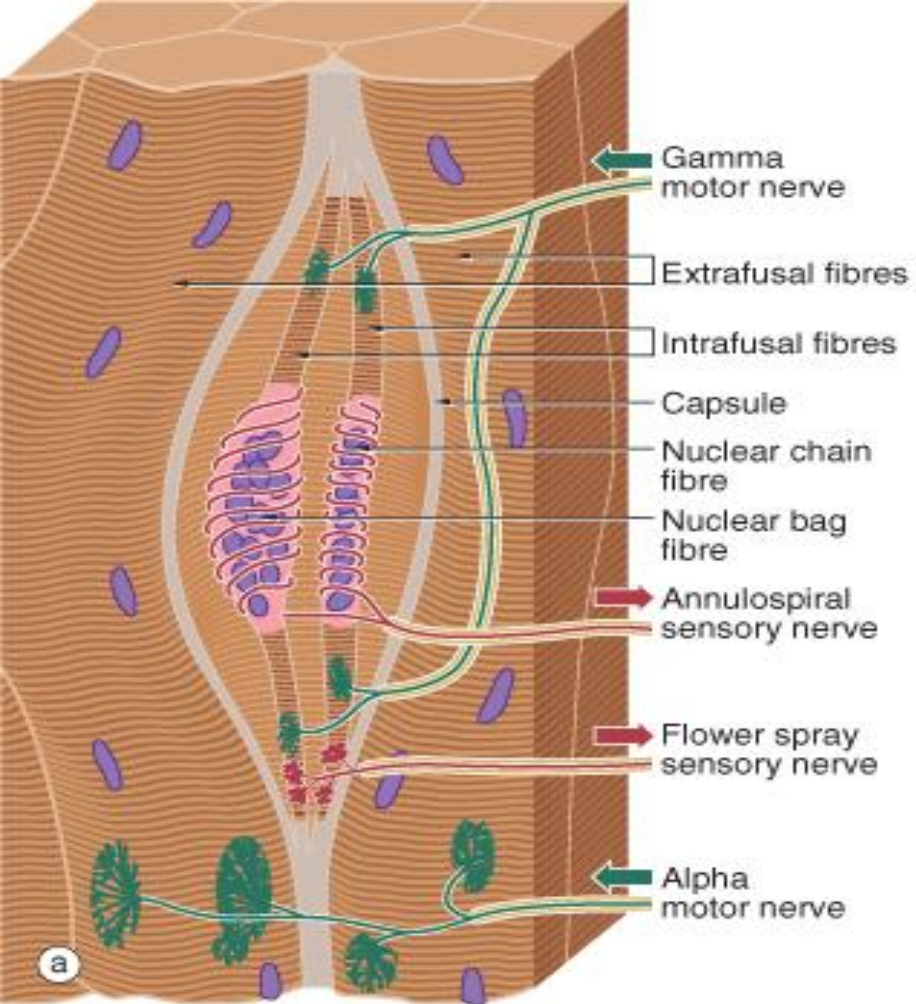


Cross-
bridge

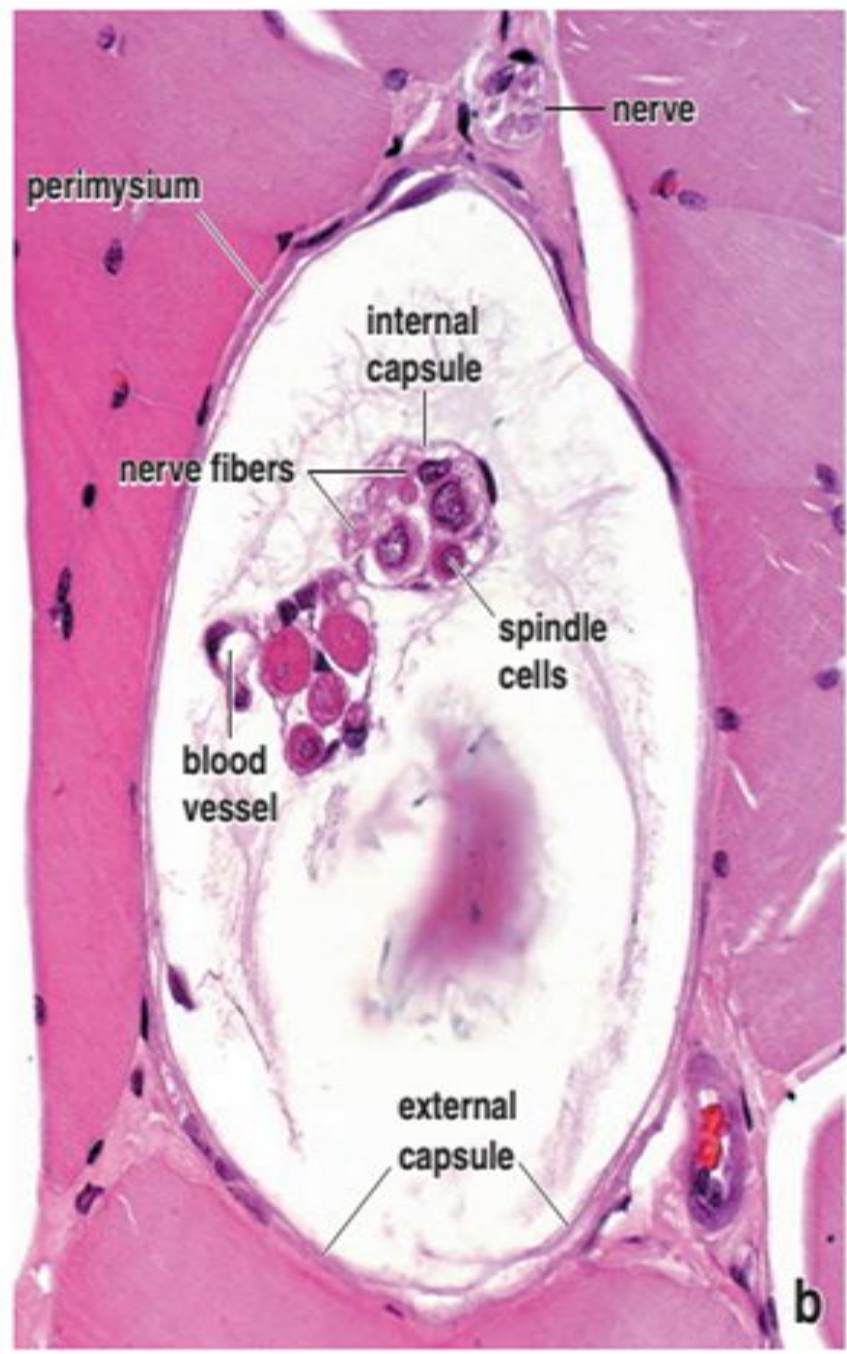
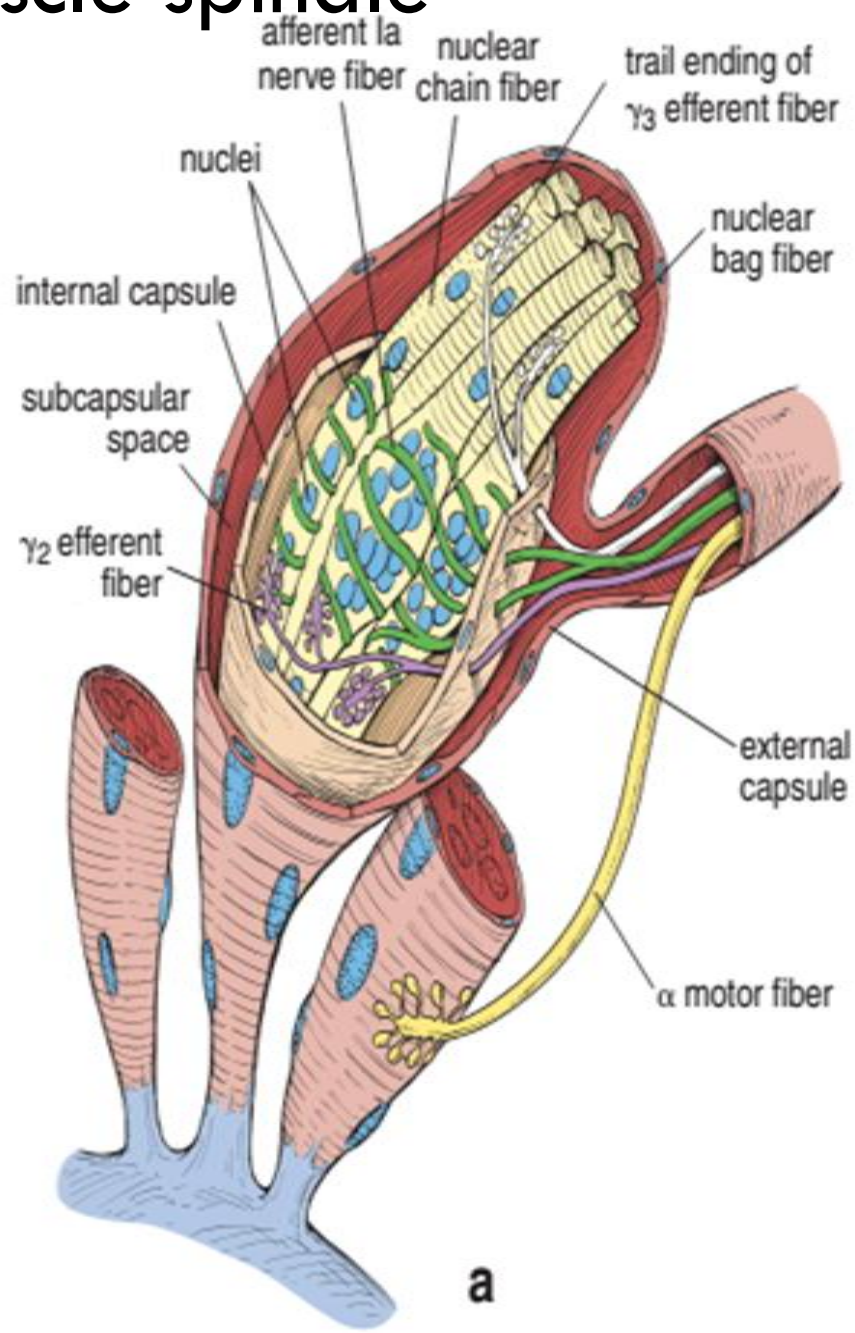


Actin



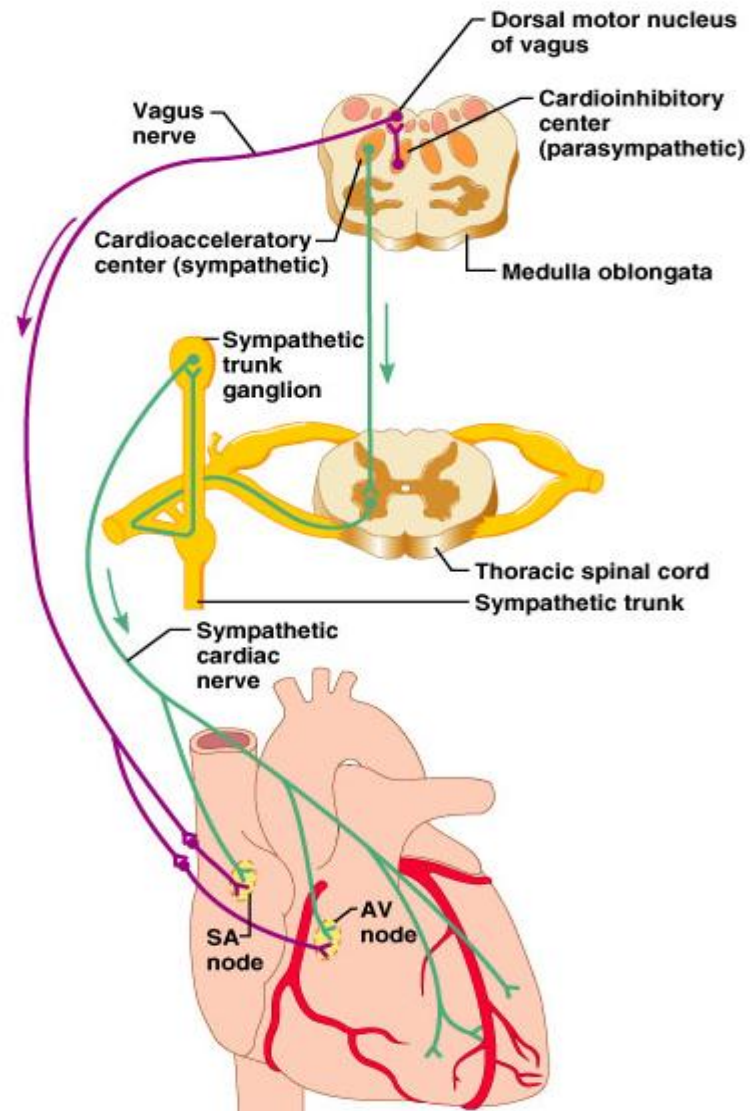
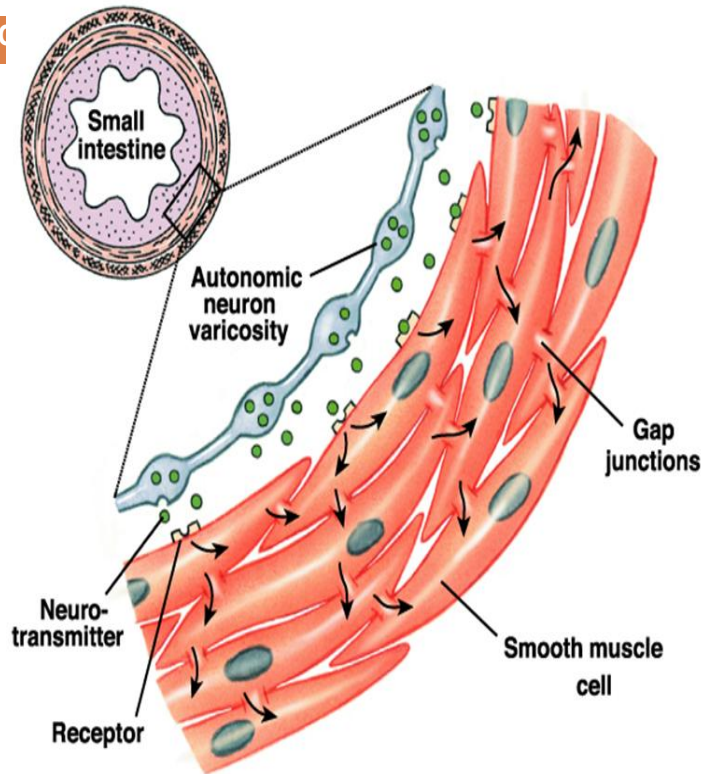


Muscle spindle



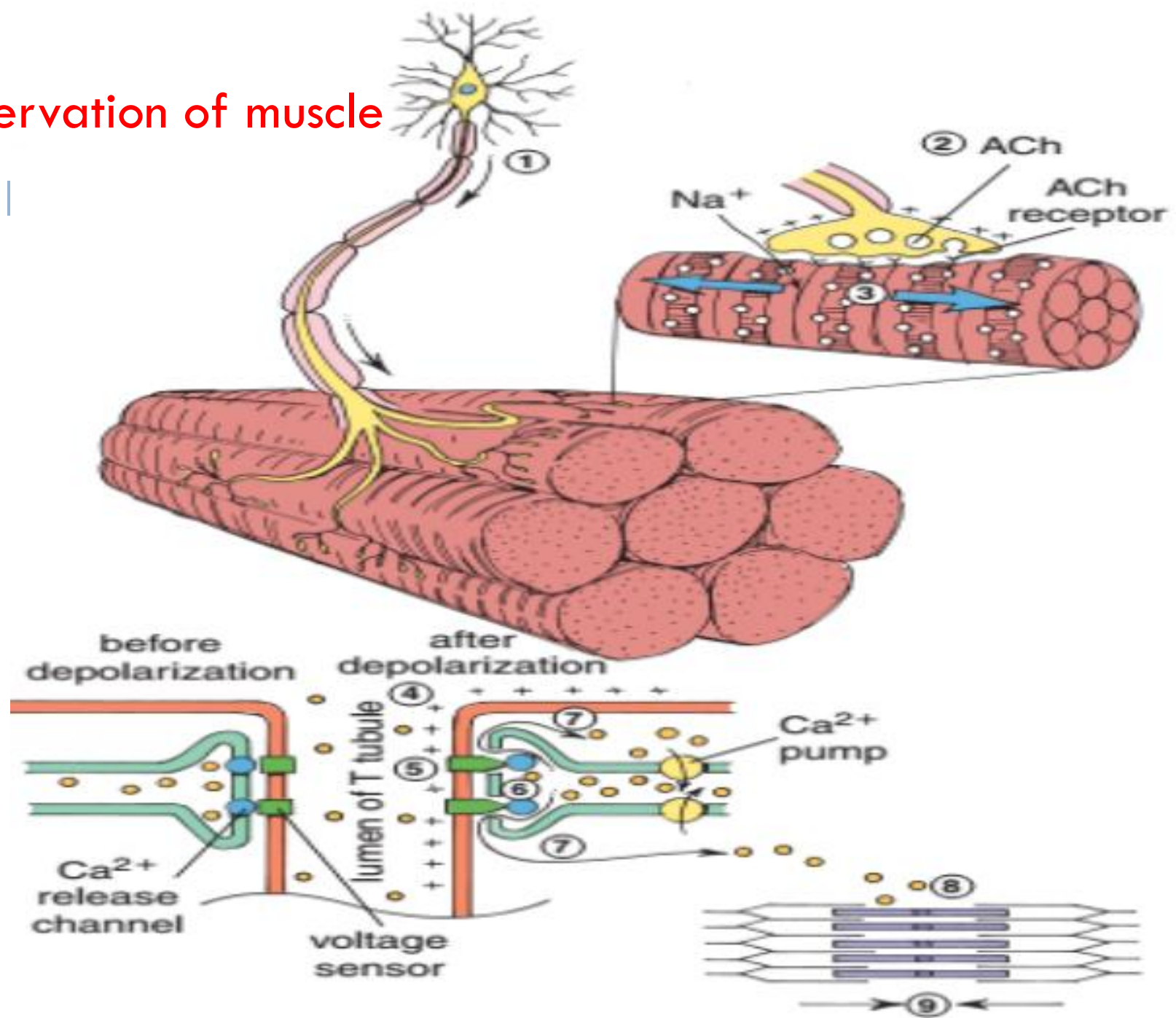
Autonomic innervation

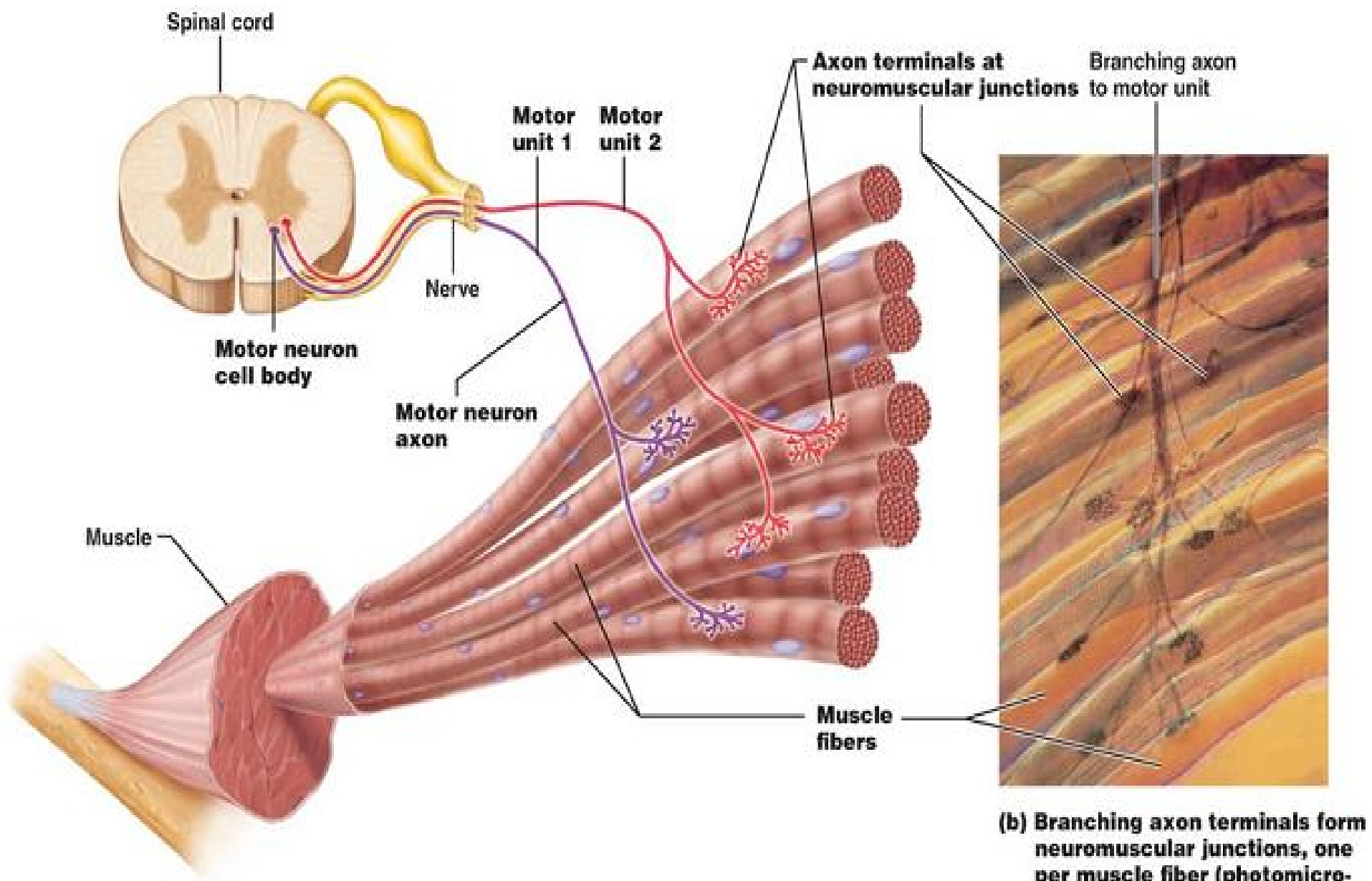
(a) Single-unit smooth muscle



- Sympathetic
 - ▣ Increases rate and force of contractions
- Parasympathetic (branches of Vagus n.)
 - ▣ Slows the heart rate

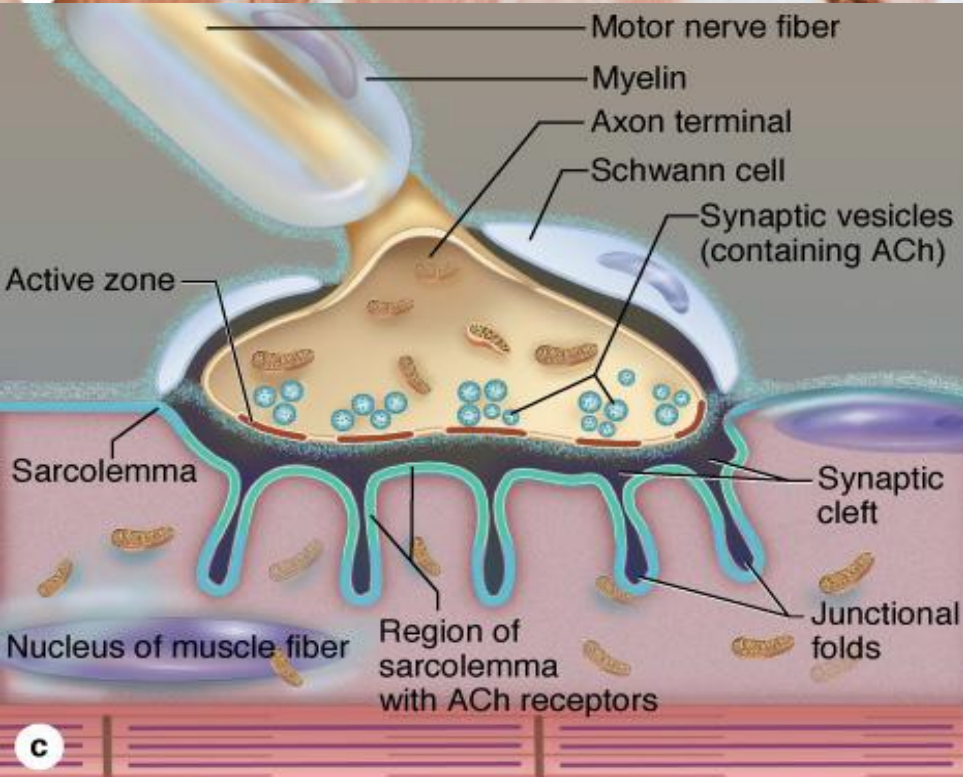
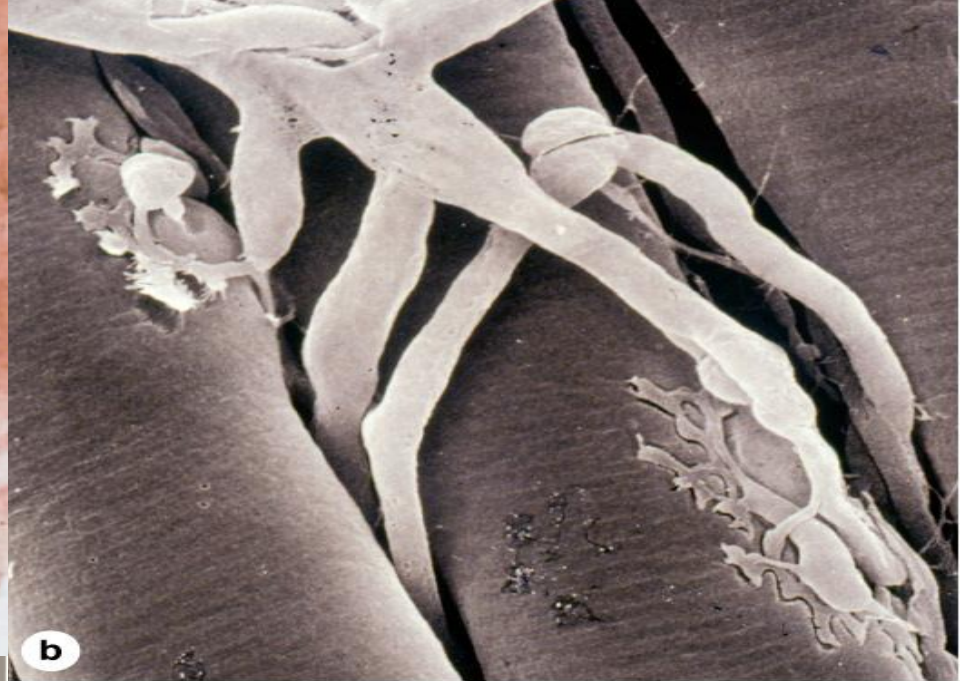
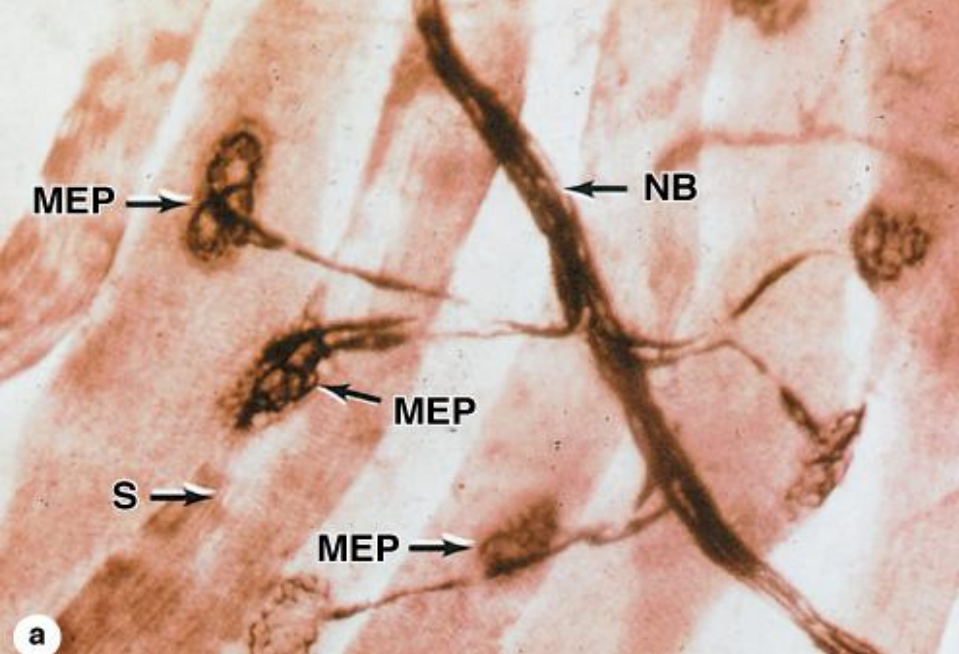
Innervation of muscle



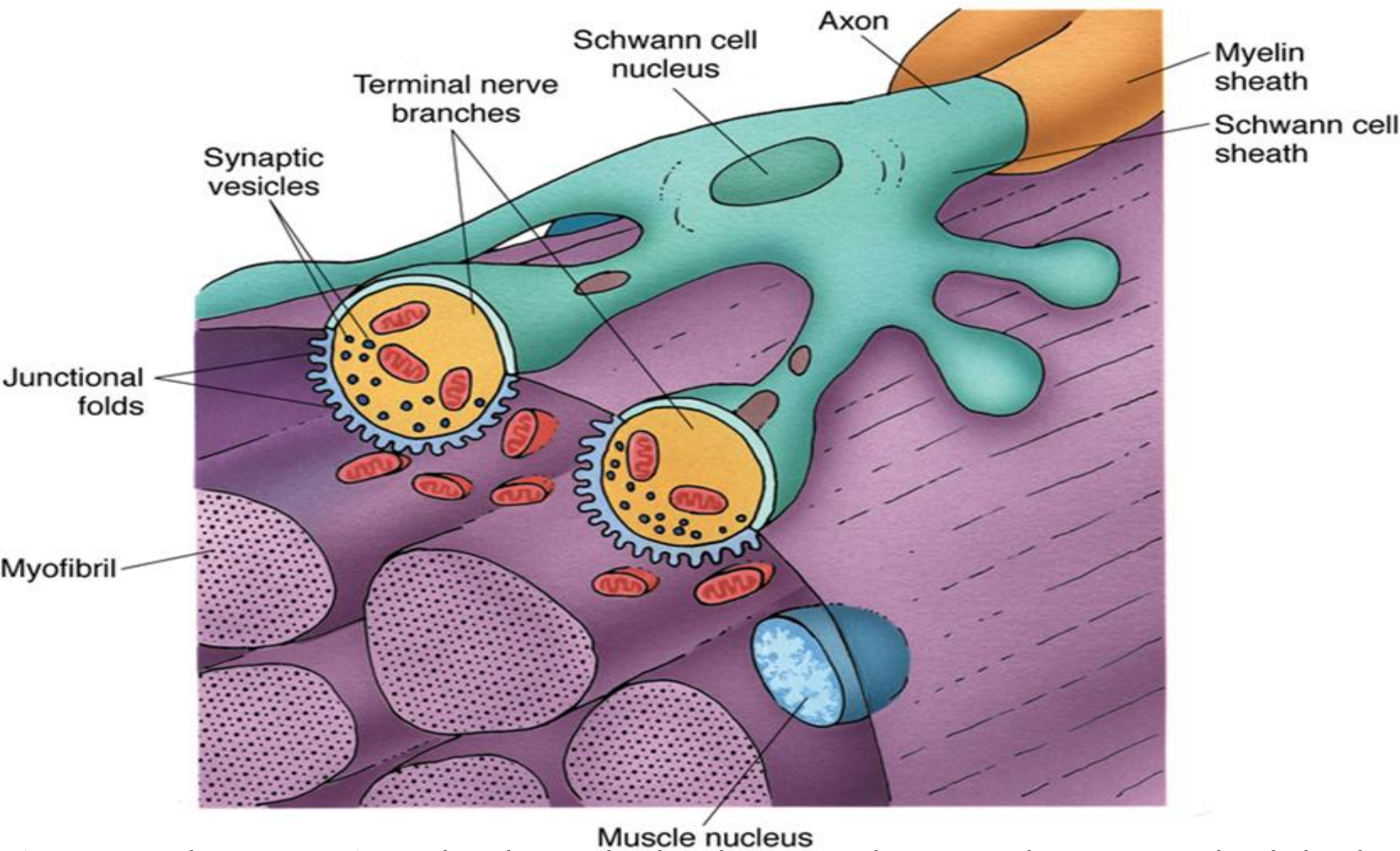


(a) Axons of motor neurons extend from the spinal cord to the muscle. There each axon divides into a number of axon terminals that form neuromuscular junctions with muscle fibers scattered throughout the muscle.

(b) Branching axon terminals form neuromuscular junctions, one per muscle fiber (photomicrograph 330 $\times$).

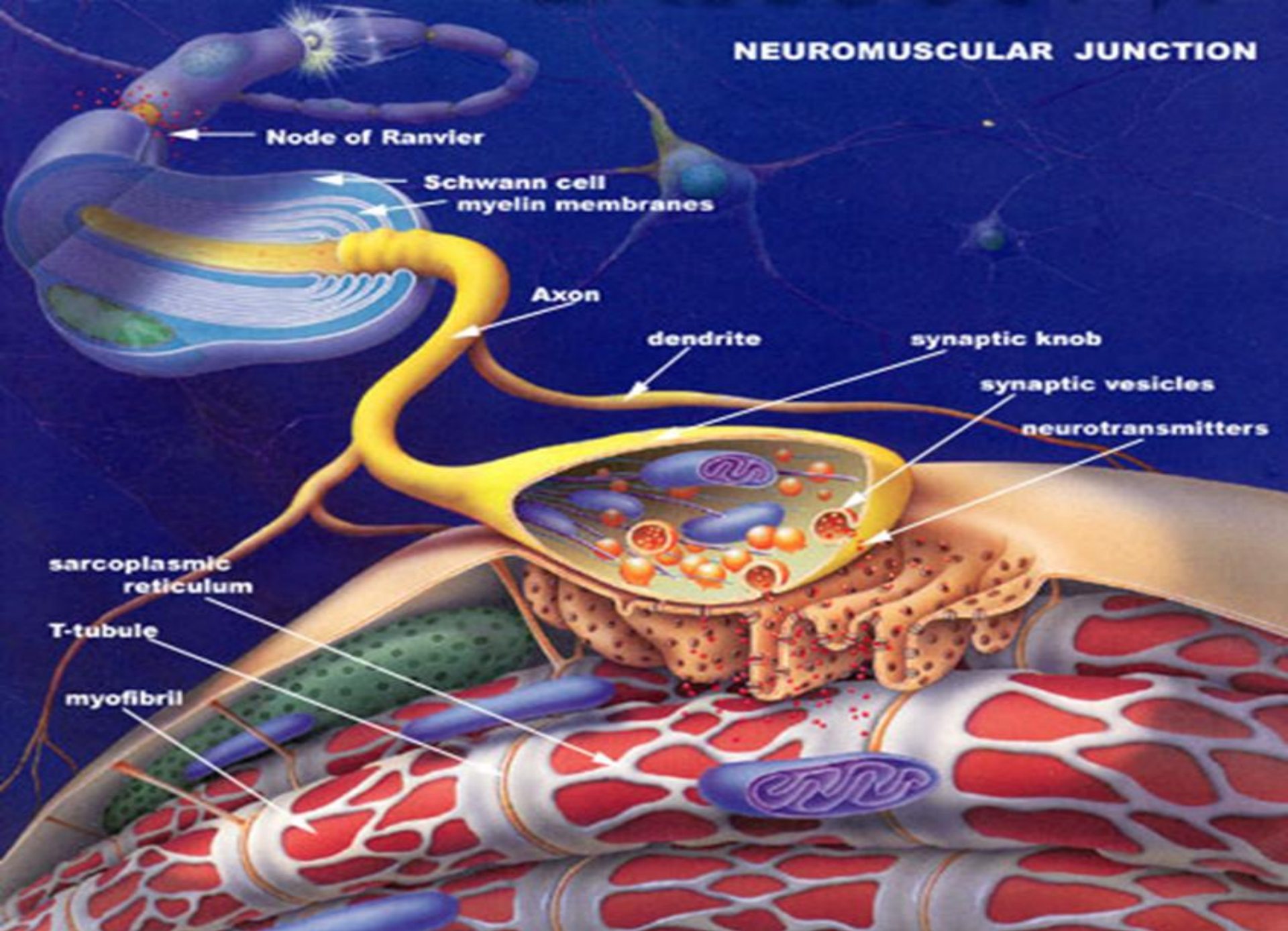


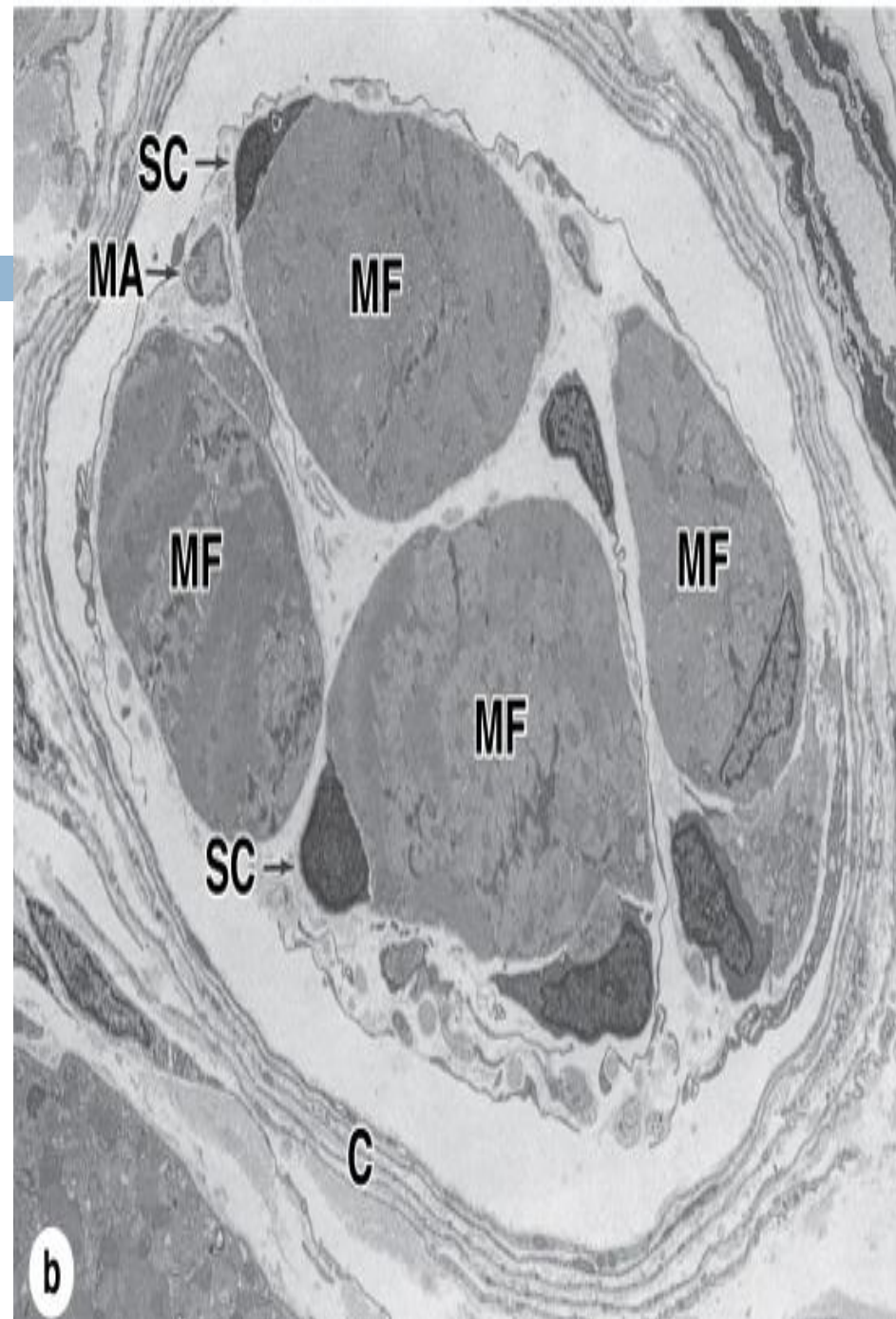
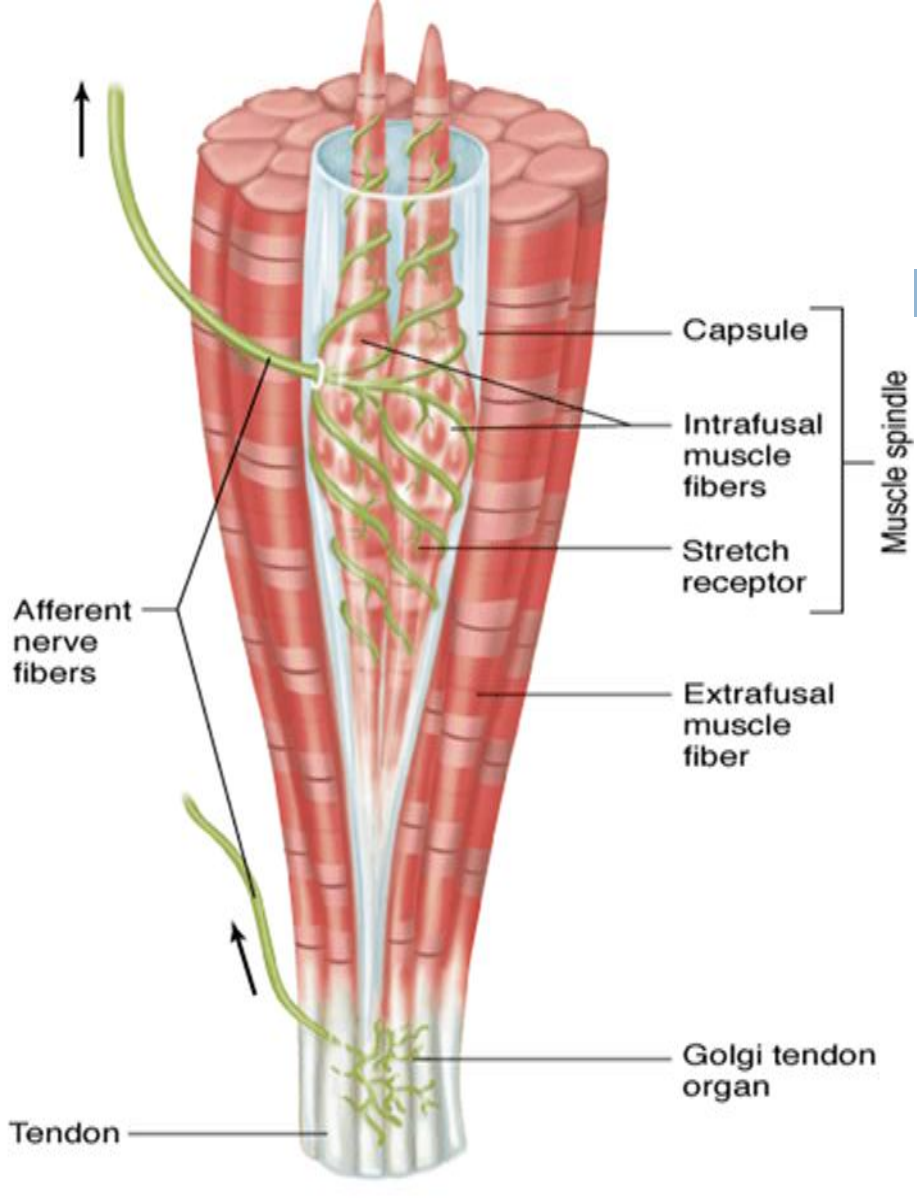
The neuromuscular junction (NMJ).



Neuromuscular junction. Note that the myelin sheath stops as the axon arborizes over the skeletal muscle fiber, but the Schwann cell sheath continues to insulate the nerve fiber. The terminal nerve branches expand to form axon terminals that overlie the motor endplates of individual muscle fibers.

NEUROMUSCULAR JUNCTION





Sensory receptors associated with skeletal muscle.

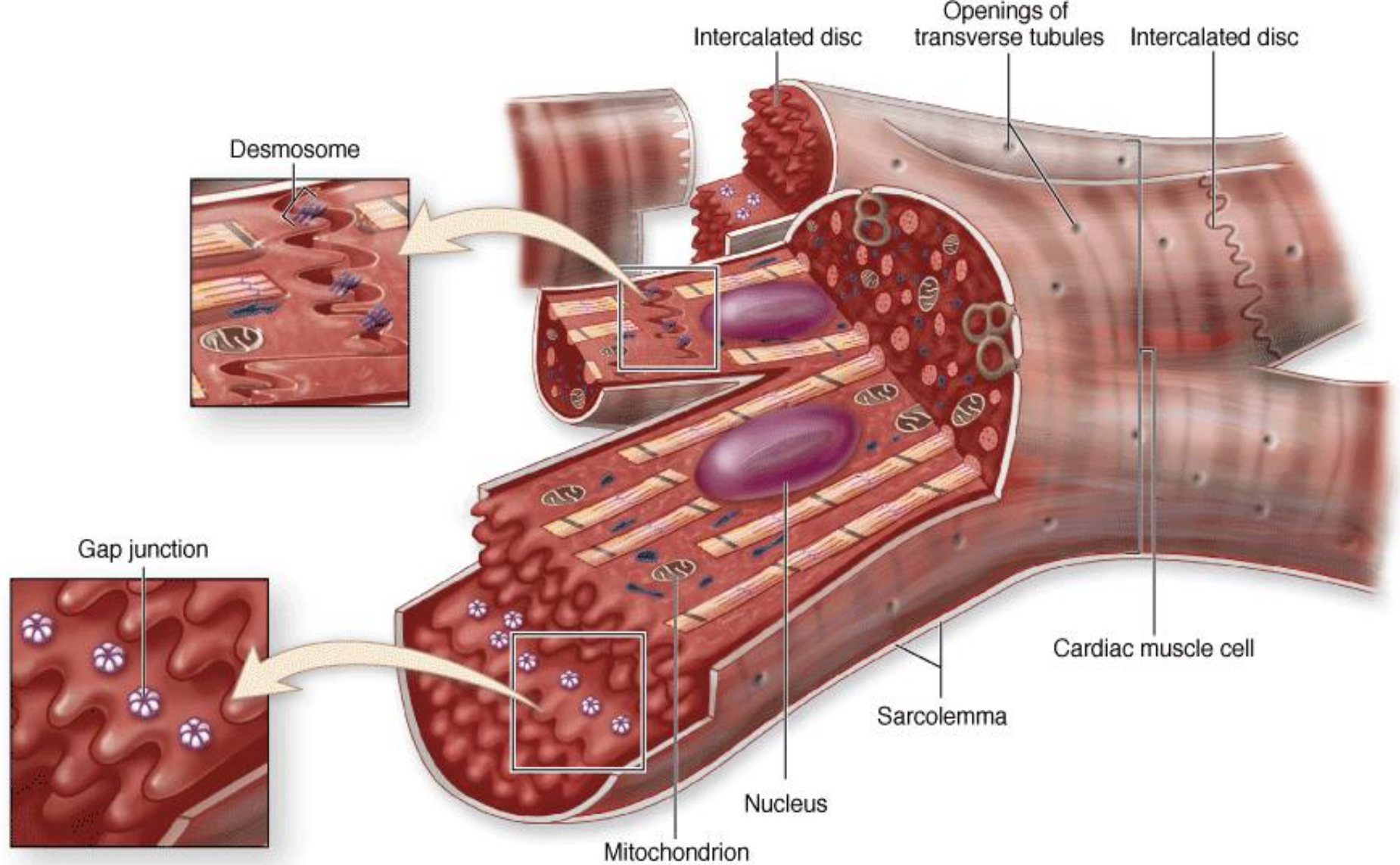
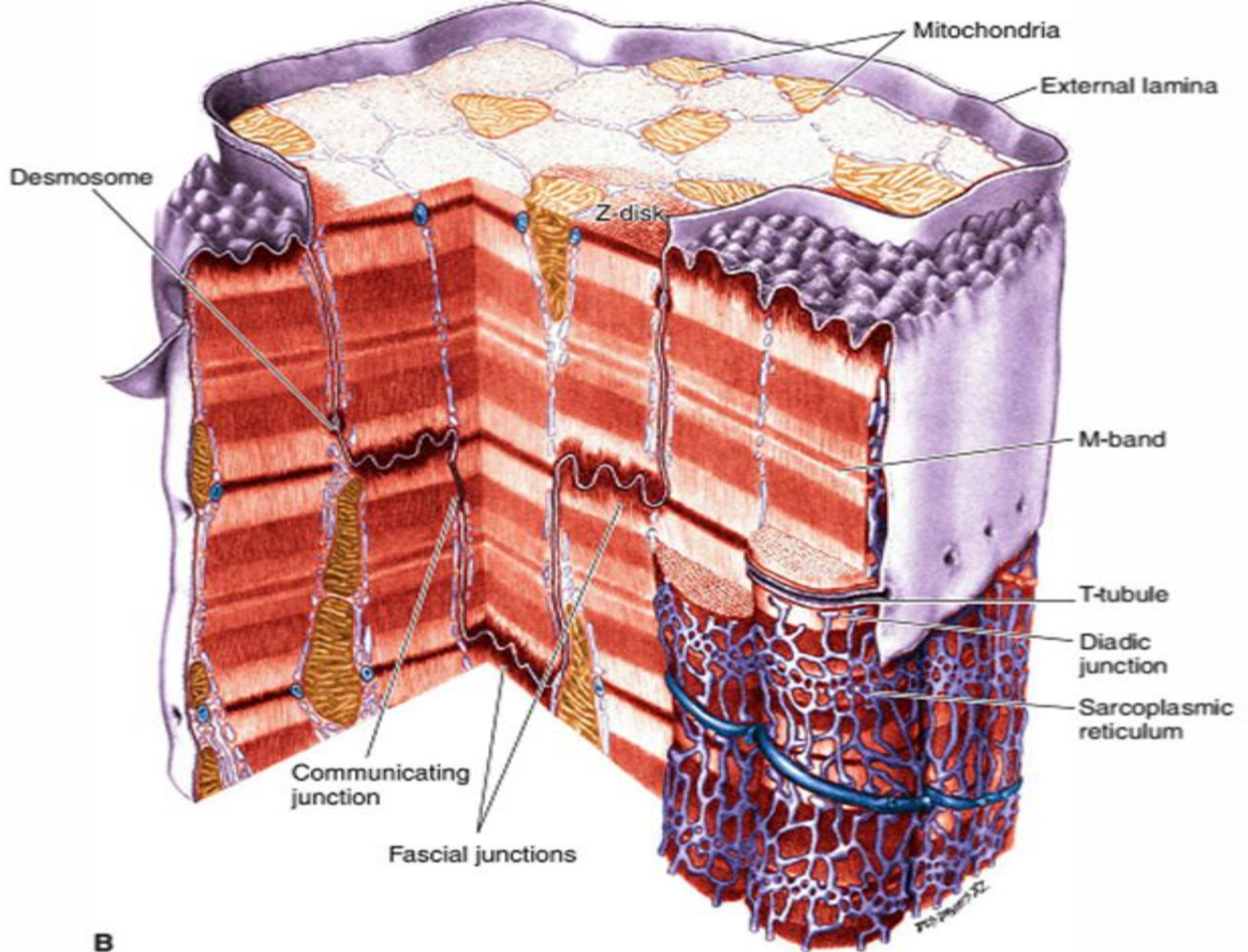
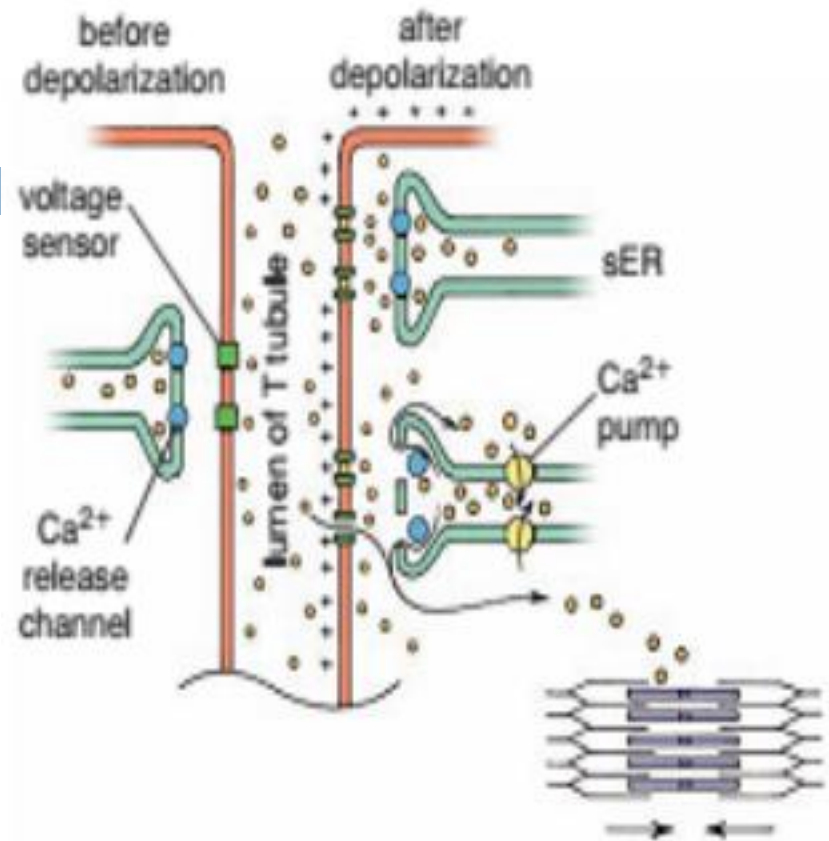
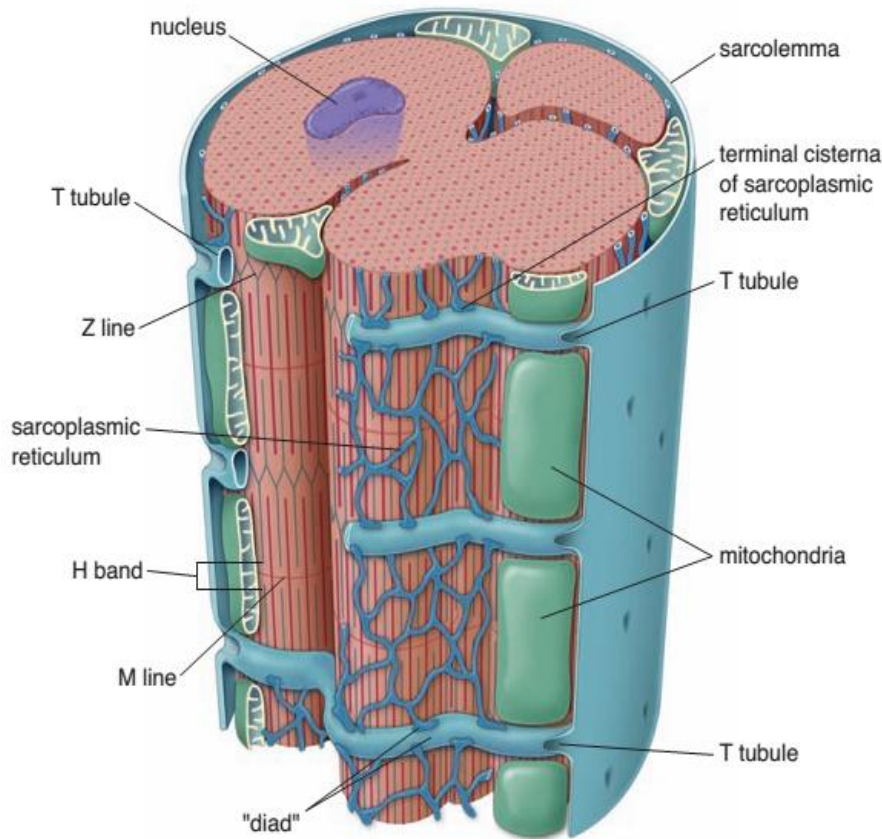
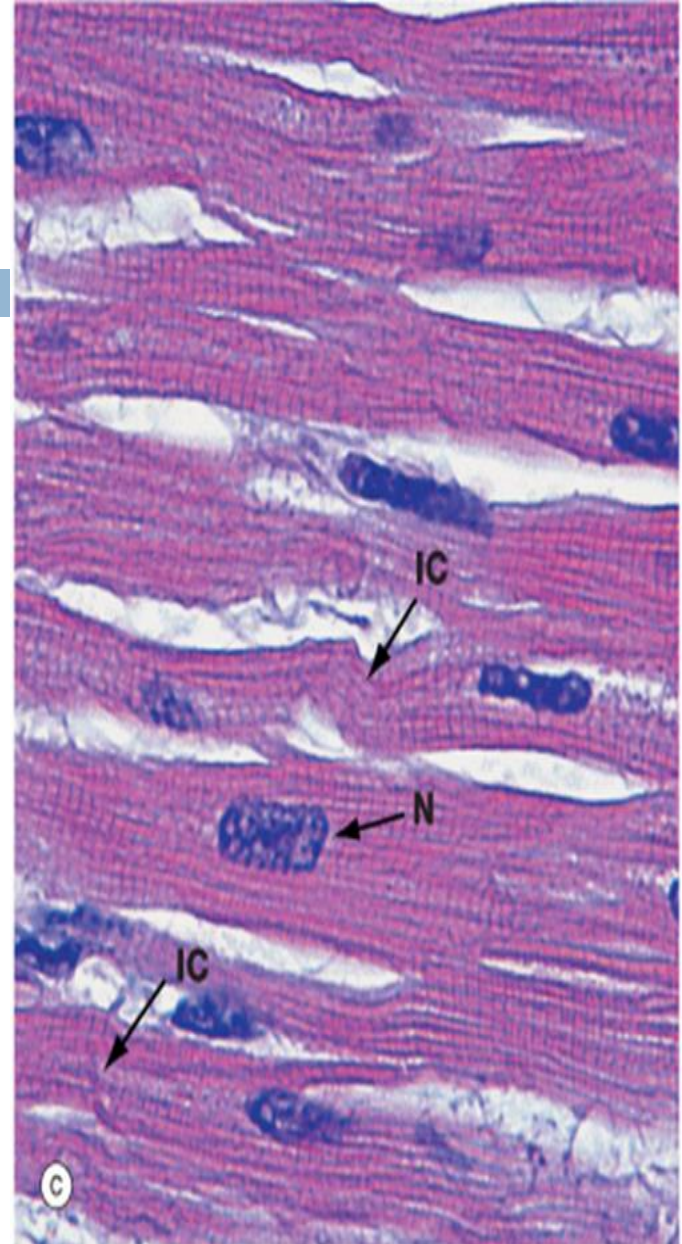
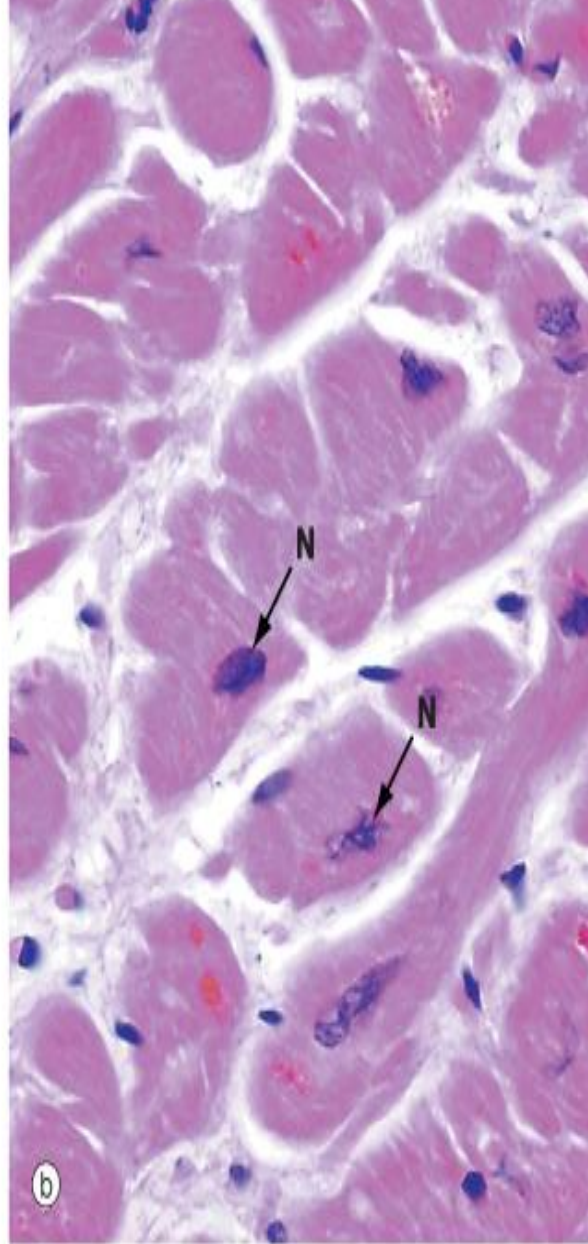
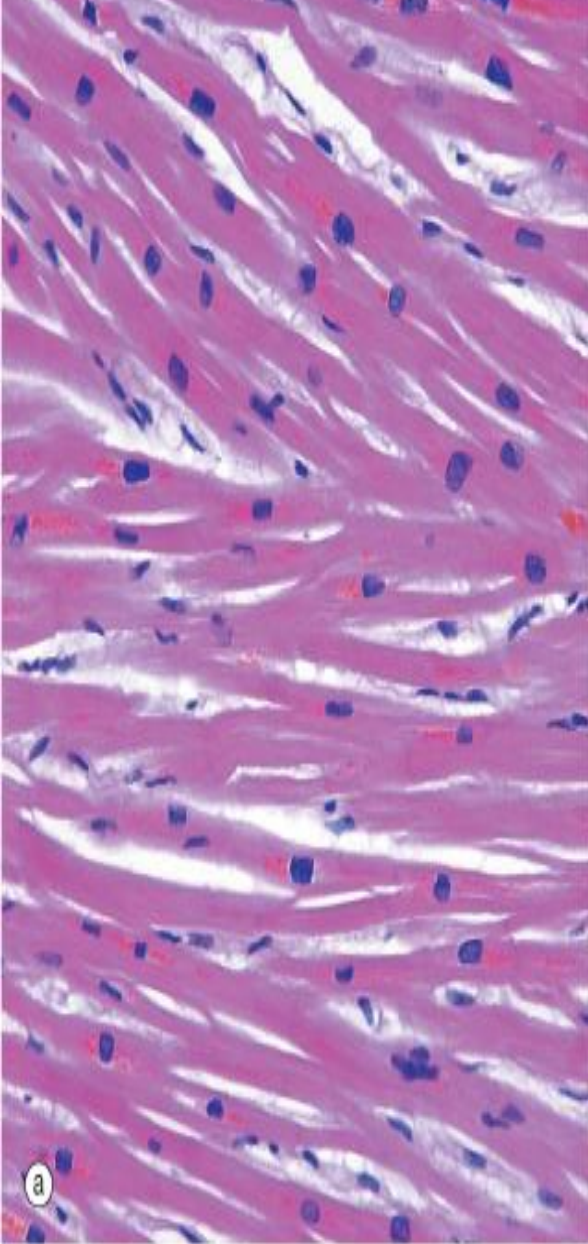


Diagram of cardiac muscle cells indicates characteristic features of this muscle type.



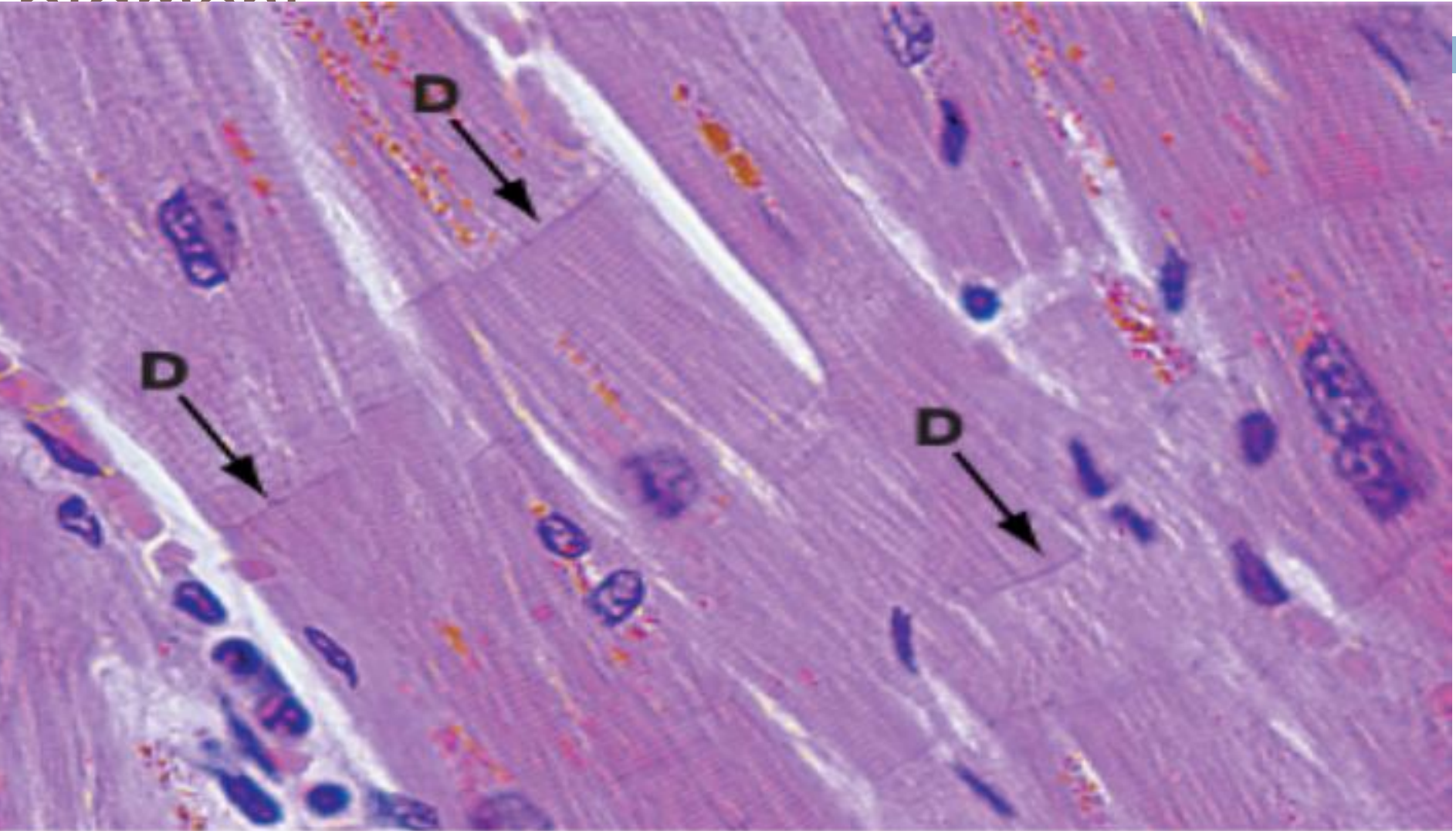


Movement of the calcium ions after depolarization of the plasma membrane in cardiac muscle. Depolarization of the T- tubule membrane activates voltage-sensor proteins that function as Ca^{2+} channels. Initially, Ca^{2+} is transported from the lumen of the T tubule through channels in voltage-sensor proteins into the sarcoplasm of the cardiac muscle (illustrated next to the upper terminal sac of the sER). Next, Ca^{2+} activates gated Ca^{2+} release channels in adjacent terminal sacs of the sarcoplasmic reticulum. This causes the massive release of sequestered Ca^{2+} from the sER into the sarcoplasm and initiates the contraction cycle.



Cardiac muscle (a) LS, H [amp] E $\times 198$ (b) TS, H [amp] E $\times 480$ (c) LS, H [amp] E polarised light $\times 480$

Intercalated disc and lipofuscin pigment



A red-brown pigment seen in these cardiac cells is termed *lipofuscin*

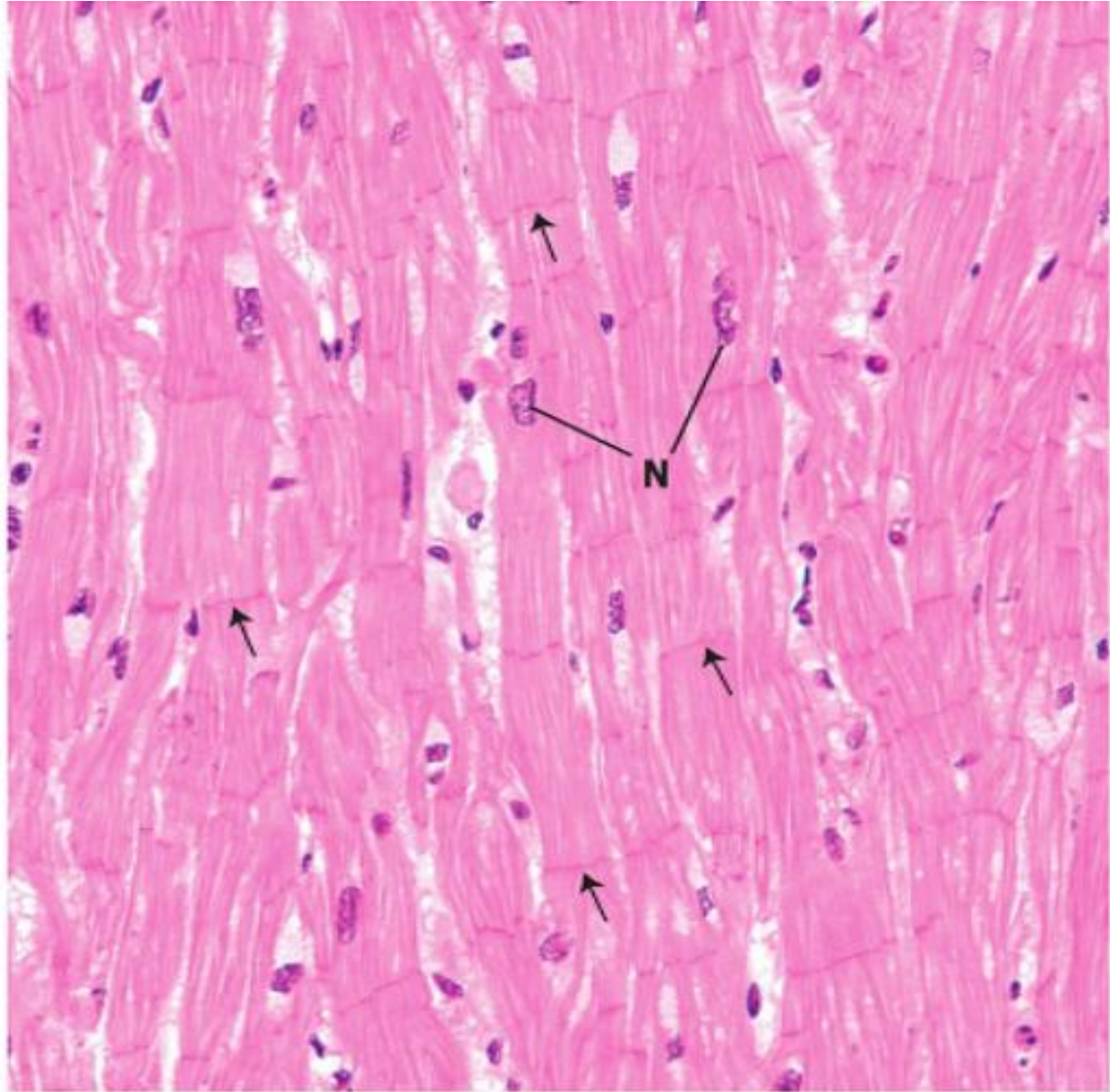


FIGURE 8.5. Light micrograph of a longitudinal section of monkey cardiac muscle fibers. N, nuclei of cardiac muscle cells; arrows, intercalated disks. Plastic section ($\times 270$).



Thank you